

Part IV

Institutional Aspect of Central Bank Policy

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Chapter 10

Monetary Policy Credibility and Time Consistency

10.1. Introduction

During the 1970s and 1980s, the general theme of monetary policy discussions was oriented more toward the selection of appropriate strategic and operational frameworks, as explained in Chapters 6 and 7, while the institutional aspects of policy implementation did not receive much attention. Only in the 1990s did institutional aspects become the subject of intensive discussions, which was partially precipitated by the emergence of the long debate concerning the formation of the European Central Bank (ECB) at the beginning of the decade. One area that received attention was how to design the right legislative framework for the existence of a central bank so that all parties mandated with implementing monetary policy could act to achieve the policy objectives. In this case, the policy goal was, typically, to achieve price stability (low inflation) in the medium–long term, while paying due consideration to supply and demand-side shocks or output performance in the near term.¹

An important lesson could be extracted from the empirical experience of policy implementation by nearly all central banks around the world, namely that inflation bias was primarily attributable to the political perspective, which sought to achieve monumental changes in a short period of time. Such conditions necessitated a legislative framework that provided central bankers independence from political interference, coupled with relatively long tenure.² In this case, longer

¹The increasing shift in policy preference towards achieving price stability was based on the belief that price stability (low inflation) was a prerequisite of sustainable economic growth and, therefore, public welfare.

²The issue of central bank independence has been around since the inception of central banks. David Ricardo voiced his support for an autonomous central bank proscribed from financing the government's budget deficit in 1824 (Fraser, 1994).

In the dictionary, independence is defined as free from outside influences, instructions or control. When applied to central bank independence, Meyer (2000) interpreted independence as free from government influence, instruction and control, both

tenures were believed to protect the central banker from political interference. Nevertheless, the status of central bank independence, in many countries, raised the question of how the institution could effectively be monitored by the public and representative members of parliament. Monitoring was required because of the risk of incompetent policymaking at the central bank, or the possibility of introducing policy that conflicted with the mandate of the central bank. Therefore, mandating central bank independence had the consequence of necessitating accountability.³

A study focusing on both institutional aspects would be pertinent considering the crucial role both aspects play in terms of improving central bank policy credibility in the eyes of the public. Policy credibility can be defined as central bank commitment to comply with the rules or policy goals articulated in a transparent manner. Simply put, therefore, policy credibility is built through consistency between what is said and what is done. In essence, for a central bank mandated with controlling inflation, credibility is a function of the difference between actual inflation and the central bank's inflation target.

Why is credibility important? Blinder (2000) stated that with credibility a central bank could reassure the public that, through the policy measures taken, the policy goals will be achieved. Therefore, higher credibility at a central bank is generally believed to enhance the implementation effectiveness of monetary policy itself because if a central bank is credible and the public have rational expectations, then delivering commitment or a statement on the planned policy to control inflation will elicit a positive public response, hence the achievement of price stability could be strived toward without sacrificing economic growth. According to Perrier and Amano (2000), a credible monetary authority could form more accurate judgments on economic capacity in terms of producing goods and services, thereby creating job opportunities without stoking concerns regarding the emergence of inflationary pressures. This is an important benefit of policy credibility, particularly when there is uncertainty concerning the output gap.

Although the important role of credibility in terms of monetary policy implementation has long been recognized, modern studies only began to appear at the end of the 1970s through the seminal works of Kydland and Prescott (1977) as well as Calvo (1978). Both studies showed that the rational expectations hypothesis has fundamental implications underlying macroeconomic policy credibility in general and monetary policy credibility specifically. One implication is that

executive and legislative. Fraser (1994), on the other hand, defined central bank independence as freedom for the central bank to implement monetary policy without political considerations.

³According to the dictionary, accountability is defined as the obligation to justify actions or decisions, being responsible. There are two opposing opinions concerning the relationship between independence and accountability. Fraser (1994) opined that accountability could help preserve central bank independence. In contrast, Meyer (2000) suggested a trade-off between independence and accountability, implying that something which increased independence would reduce accountability and vice versa.

monetary policy inconsistency would lead to inflation exceeding its target, or an inflation bias, because the central bank is seeking stronger economic growth. Therefore, if the central bank is not pre-committed to achieving the inflation target, then the subsequent trends that transpire will be inconsistent with monetary policy implementation and a lack of credibility will be conveyed through the policy statement of the central bank.

In this chapter, aspects of central bank credibility will be discussed along with several measures that could be taken to improve credibility. The following two sections will describe several conceptual dimensions and present a survey of the literature, as well as the theoretical models linked to policy credibility, specifically in terms of monetary policy inconsistency. The fourth part will review the empirical observations and several important related issues, such as measures of the credibility index and time inconsistency. An empirical assessment is presented in the fifth section regarding monetary policy credibility in Indonesia, followed by the concluding remarks. It is also important to note that improving central bank policy credibility is fundamentally linked to the development of institutional aspects, such as policy independence, accountability and transparency, as discussed in Chapters 11 and 12.

10.2. Conceptual Dimension of Policy Credibility

On a pragmatic level, the issue of credibility can be observed from two perspectives, namely policy credibility and target credibility. Policy credibility relates to how consistent public expectations are of policy implementation by the central bank compared to the policy plan previously announced by the central bank. This depends on whether policy implementation by the central bank deviates from the previous statement due to outside interference or economic changes. Meanwhile, target credibility relates to how consistent public expectations are of achieving the future policy target compared to the target announced previously by the central bank. This depends on the level of policy credibility and whether the economic structure, including the public response, facilitates achievement of the previously announced target.

Therefore, policy and target credibility are fundamentally determined by at least three aspects: (1) the public response reflected in the formation of expectations; (2) central bank capacity to maintain and commit to the policy strategy over time; and (3) the structure and development of the economy that facilitates sound policy transmission mechanisms. For example, if the public feel the central bank is unable to implement policy or could implement inconsistent policy, the policy will probably fail. The more often the central bank fails in its policy implementation, the lower its credibility will be in the eyes of the public. Consequently, it will become increasingly difficult for the central bank to implement policy, which is often referred to as the “credibility problem.”

10.2.1 Causes of the Credibility Problem

Analytically and practically, there are several causes of the credibility problem on the policymaking side as follows:

(1) Time inconsistency

Time inconsistency occurs when the policy authority faces a dilemma in determining the policy preferences in relation to the time dimension. For instance, the policy authority pledges to implement policy to subdue inflationary pressures in a given period. After the economic agents respond to the decision by setting nominal wages based on the expectation of low inflation, however, the policy authority could be tempted to tolerate higher inflation in the subsequent period in order to increase output and allow the economy to absorb more labor. This could occur if the policymakers are confronted with the dilemma of controlling inflation or lowering unemployment. Implementing such a policy would quickly erode the credibility of the policy authority because when setting the same policy in the subsequent period the economic agents will be less inclined to trust the authority and will anticipate conditions based on previous experience by setting higher nominal wages.

(2) Internal inconsistency

Internal inconsistency occurs if the policy authority faces a policy dilemma related to sectoral interests. In general, the issues relate to a lack of policy coordination, including policies in a certain policy authority's domain as well as policies between the policy authorities, which leads to contradictory policies and, therefore, the credibility problem. An example would be deflationary policy coupled with an unrestricted increase in the budget deficit, which could occur if policymakers instituted tight money policy together with an excessive budget deficit policy. Another example is the inconsistent implementation of an economic reform program or a consistent program but with asymmetrical implementation.

(3) Asymmetric information

The third source of the credibility problem is asymmetric information or incomplete information concerning the policy authority. Economic agents may not be aware that the policy authority is seriously pursuing a deflationary policy, thus when the policy is announced, the economic agents who are not convinced of the policy authority's commitment require time to learn and convince themselves that the policy authority will really implement the desired policy. Such conditions typically occur in developing countries, where the policy authority tends to be inconsistent and implement inconsistent policies that confuse the economic agents. Consequently, the economic agents are not sure about the goal of the policy. Furthermore, as the economic agents do not have access to symmetric information, they are also unable to monitor what the policy authority is actually trying to achieve. Therefore, it becomes difficult to build a solid reputation that the policy authority is seriously pursuing a particular economic policy.

(4) Stochastic economic shocks

Economic shocks that are difficult to predict (stochastic) could cause a policy program to fail in the achievement of its target despite sound design and consistent implementation because such shocks could create off-target public expectations concerning an economic indicator set by the policy authority. In the context of economic stabilization policy, the various shocks that appear

could lead to an inaccurate economic stabilization program. If the economic agents subsequently assume an inaccurate economic stabilization program, their expectations of the program will be low and the program could reasonably run into problems or even fail.

(5) Political uncertainty

The presumption among economic agents that the policymakers are unable to implement an economic policy package due to a lack of domestic political support could also exacerbate the credibility problem, which could occur if the government is a coalition of various political parties or if the government's legitimacy is questioned by the public. Consequently, even if the economic stabilization policy package issued by the government is sound, the lack of political support, coupled with vested policy-making interests, could trigger the credibility problem.

In practice, several of the factors mentioned above combine to influence the problem of policy-making credibility. Consequently, it is nearly impossible to measure the magnitude of the credibility problem caused by each respective aspect, especially in terms of certain economic packages or programs. In general, credible programs are assumed to succeed, while programs lacking credibility tend to fail. Nevertheless, the sources of the failure are difficult to separate individually.

10.2.2. Effort to Improve Credibility

Improving credibility requires serious efforts by the policy authority. The various determinants of the credibility problem listed above demand vigilance in terms of time and sector inconsistency, information availability, anticipation of the elemental ideas, as well as process assurance. Several efforts can be applied to improve policy credibility.

(1) Using a rule

Using a specific policy rule is thought to prevent the policy authority from "deceiving" the markets. The use of a rule is expected to reduce uncertainty concerning future policies and, therefore, stabilize expectations and macroeconomic variables. In practice, an example of a rule is through commitment to a fixed exchange rate. By implementing a fixed exchange rate system, the central bank is expected to alleviate inflationary pressures on the domestic currency due to external pressures and simultaneously ease inflation bias because the public understands the consequences of applying a rule to monetary policy. Therefore, if the public and policymakers are jointly committed to a specific rule, then its use will help overcome the credibility problem. Conversely, if commitment to a rule is not fulfilled, the relevance of the implemented policy would be undermined or even become counterproductive.

Operationally, a rule could be applied using an operational target or policy response that adheres to rules for certain variables, for example based on monetary aggregates (Friedman rule and McCallum rule) or based on the interest rate (Taylor rule), as discussed in Chapter 7.

(2) Building reputation

The academic community has explored various solutions to the time inconsistency problem. One solution was proposed by Barro and Gordon (1983), namely by building reputation. They explained that the time inconsistency problem emerges because the public knows that the central bank has an incentive to accommodate measures to achieve low inflation (to anchor low public inflation expectations) and subsequently tends to shock the public with high inflation to stimulate economic growth or reduce unemployment. Nonetheless, if the central bank seriously and constantly strives to build its reputation (which typically requires a protracted period), by meeting its commitments, the public will gradually begin to trust the actions taken by the central bank. Consequently, if the central bank announces its commitment to maintain low inflation, the public will trust the move and keep their inflation expectations low. Oppositely, by setting high inflation, the central bank will damage its reputation and find it more difficult to catalyze economic growth in future periods.

It is important to note that when building reputation, the policy authority must signal its seriousness to the economic agents in terms of economic policy implementation. In many cases, this seriousness is evidenced by bold policy measures as shock therapy with significant implications on the workings of the economy. In addition, an appropriate policy sequence is required for a policy to be sustainable. For example, a microeconomic adjustment and institutional change are required before macroeconomic policy is implemented. If the policy sequence is appropriate, however, the economic agents will see that any misalignment would be detrimental to the policy authority itself and, therefore, they trust that the policy authority will not adjust the economic policy sequence.

(3) Delegation of authority

Concerning efforts to improve policy credibility through independence, one practical solution was also proposed by Rogoff (1985) through the delegation of authority. Based on the principal-agent rule, Rogoff stressed that the reigns of central bank leadership should be delegated to inflation-averse or inflation-conservative individuals rather than other types of economist. By delegating authority, a mechanism to guarantee monetary policy implementation consistent with the hopes of the public is expected.⁴ For example, if the central bank is given a high level of independence, the central bank is also expected to protect monetary policymaking from political processes. To that end, Alesina and Gatti (1995) argued that central bank independence could reduce dynamic influence and political uncertainty on output.

⁴After the seminal theoretical contribution of a “conservative central bank” (Rogoff, 1985), the literature on independence was enriched by other studies, including “insulation from political uncertainty” (Alesina & Gatti, 1995) as well as “optimal contract” (Walsh, 1995). Meanwhile, a prominent theoretical argument regarding accountability is contained in “democratic accountability” (Eijffinger, Hoeberichts, & Schaling, 2000).

In addition, the central bank could join a monetary union and apply a fixed exchange rate system, thereby submitting to the monetary policymaking of the monetary union. By capitulating to the exchange rate policy and monetary policy of a large and credible central bank, the credibility of the policy authority itself could also be improved. This is an effective method for relatively small countries or those suffering from comparatively low policy credibility.

(4) Contract determination

Walsh (1995) opined that if inflation bias is present, namely that if a central bank is tempted to precipitate economic change that triggers inflationary pressures, this may be due to a lack of incentives. Therefore, an optimal contract must be designed for the central bank to prevent, or at least reduce, potential inflation bias. In this context, the government could offer a contract, for instance through performance-based salaries for central bankers, in accordance with achieving the inflation target or another economic variable at a certain level, for example, money supply growth. In this case, central bankers would be rewarded if inflation is successfully controlled but also penalized in failure or a lack of accountability. Such an optimal incentive structure in terms of controlling inflation is congruent with the current trend of setting an inflation target as the overriding goal of monetary policy.

(5) Political support for the rule of law

The benefits of an economic policy must ultimately be redistributed to the wider community. If the policy authority does not enjoy the support of society at large, the economic policies taken will not run smoothly. In general, policy authority credibility depends upon support from the political party in power, legitimate in the eyes of the public. Such political support is typically obtained when a new government administration is elected. The public tends to accept the policies of the new administration even though such policies could spur economic pressures in the near term. Nevertheless, support can quickly turn to opposition in the event of an extended recession, which would force the policy authority to adjust the contents of the economic policy package.

Meanwhile, laws applicable to policymakers can also be viewed as an appropriate measure to improve policy credibility. This may be in the form of state laws contained in the constitution. Furthermore, unwritten rules can also apply, such as social norms or social contracts that have a strong legal foundation. In this regard, it is believed that standard behaviors or ethics when serving the public may be unwritten but, if acceptable to the public, could have a strong effect.

Based on that rationale, the importance of commitment through a rule-based policy response, reputation building or the delegation of authority is the primary aspect underlying policy credibility oversight. This is also based on the recommendations of several research papers that the central bank should adopt a monetary policy framework which prioritizes good governance, for instance inflation targeting. Through inflation targeting, as discussed in Chapters 8 and 9, a policy

framework can be built that orients central bank discipline and commitment to achieve the inflation target in the long run, while simultaneously improving monetary policy credibility in the eyes of the public.

10.3. Theoretical Model Concerning the Policy Credibility

Problem: Time Inconsistency

The model of Kydland and Prescott (1977) and further development by Barro and Gordon (1983) represent two seminal references in terms of policy credibility theory. Furthermore, both models underlie the theories of independence and accountability regarding policy transparency to be discussed in Chapters 11 and 12.

10.3.1. *Kydland and Prescott's Model*

Evidence for the problem of time inconsistency was first proposed by Kydland and Prescott (1977). In their seminal work, Kydland and Prescott analyzed the time inconsistency problem to show economic policy formulation that generally relies on a rule is better than relying on discretion. Kydland and Prescott used a dynamic game theory approach, à la Stackelberg, between the policymaker (leader) and markets (followers). They showed that although policymakers can proclaim an optimal policy and the markets may trust the claim, there is the possibility that in subsequent periods the policy implemented is no longer optimal if there is a change from the original plan. A change in policy previously considered optimal over time is known as time inconsistency and, therefore, a policy measure relying on discretion is considered suboptimal or failing to maximize public prosperity.

According to the approach, the two periods to determine the policy and social objective function of the policy are assumed as follows:

$$S = S(x_1, x_2, \pi_1, \pi_2) \quad (1)$$

where, S , x , and π represent the policymakers' preferences, economic agents' decision variable and policymakers' (instrument) decision variable.

If, in the economic decision-making process, the economic agent is assumed to consider the policies formulated by the authority, then:

$$x_1 = x_1(\pi_1, \pi_2) \quad \text{and} \quad x_2 = x_2(x_1, \pi_1, \pi_2) \quad (2)$$

Seeking an optimal intertemporal solution, the decision of both policy instrument variables is taken in period 1. Therefore, by substituting the values of x_1 and x_2 , the objective function of the policymaker (1) can be expressed as follows:

$$S(\cdot) = S(x_1(\pi_1, \pi_2), x_2(x_1(\pi_1, \pi_2), \pi_1, \pi_2), \pi_1, \pi_2) \quad (3)$$

Therefore, the first-order conditions for optimal decisionmaking based on a rule, namely π_1^* and π_2^* are as follows:

$$\begin{aligned} \partial S(\cdot) / \partial \pi_2 &= [\partial S / \partial x_1 \cdot \partial x_1 / \partial \pi_2 + \partial S / \partial x_2 \cdot \partial x_2 / \partial x_1 \cdot \partial x_1 / \partial \pi_2] \\ &+ [\partial S / \partial x_2 \cdot \partial x_2 / \partial \pi_2 + \partial S / \partial \pi_2] = 0 \end{aligned}$$

Thus: $\partial S / \partial x_2 \times \partial x_2 / \partial \pi_2 + \partial S / \partial \pi_2$ (4)

$$+ \partial x_1 / \partial \pi_2 [\partial S / \partial x_1 + \partial S / \partial x_2 \times \partial x_2 / \partial x_1] = 0$$

Meanwhile, for discretion-based policy, the policymaker in period 1 determines π_1^* and π_2^* in the same way as above but in period 2, π_1^* and π_2^* refer to the value already realized. Consequently, in the second period, the policymaker will take another decision based on the following conditions:

$$S = S(x_1, x_2, \pi_1, \pi_2) \quad (5)$$

with

$$x_2 = x_2(x_1, \pi_1, \pi_2); x_1 = x_1, \text{ dan } \pi_1 = \pi_1^* \quad (6)$$

Thus, the first-order condition for period 2 with discretion-based policy π_2^{**} is:

$$\partial S / \partial x_2 \cdot \partial x_2 / \partial \pi_2 + \partial S / \partial \pi_2 = 0 \quad (7)$$

From the two solutions above, the first-order condition with discretion will only be the same as the first-order condition for the rule if $\partial x_1 / \partial \pi_2 [\partial S / \partial x_1 + \partial S / \partial x_2 \times \partial x_2 / \partial x_1] = 0$. In reality, such conditions are difficult to find and therefore, the solution for a rule (π_1^*, π_2^*) will be different to the solution for discretion (π_1^*, π_2^{**}).

As the solution for a rule (π_1^*, π_2^*) maximizes the intertemporal utility, the result is optimal. Meanwhile, discretion-based policy (π_1^*, π_2^{**}) will not be optimal because discretion-based policy in period 2 would not consider the impact of the decision taken in period 1 on (x_1, π_1).

Kydland and Prescott provided a graphical illustration concerning the application of the monetary policy time inconsistency problem in terms of a dilemma, linked to the trade-off between inflation and unemployment. In this case, the relationship between inflation and unemployment is assumed to follow the Expected Augmented Phillips Curve (EAPC) as follows:

$$u_t = u^* - \alpha(\pi_t - \pi^e) \quad (8)$$

where u^* is the natural rate of unemployment. The exogenous parameter α is positive (>0), while the inflation expectations for period t are formed in period $t - 1$. As inflation exceeds the expectations, real wages decline, thereby stimulating

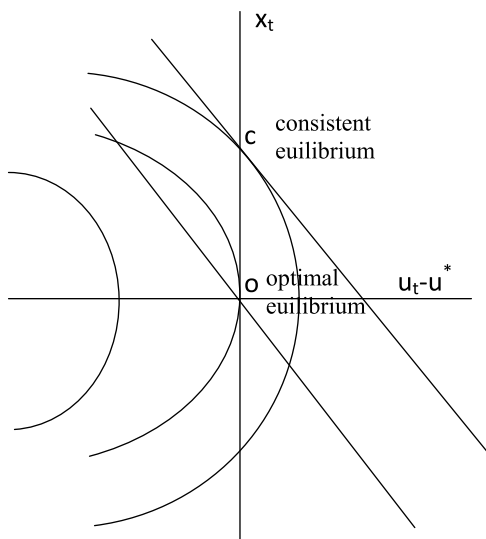


Fig. 10.1: Indifference Curve of the Time Inconsistency Problem.

Source: Kydland and Prescott (1977).

demand for labor (reducing unemployment). Meanwhile, the objective of the policymaker is to maximize the objective function:

$$S(u_t, \pi_t) \quad (9)$$

where S reached maximum at the point where $\mu_t < \mu^*$ and $\pi_t = 0$. Specifically, the formal form of S is represented by the quadratic function⁵:

$$S = (u_t - ku_t^*)^2 + \gamma\pi_t^2 \quad (10)$$

Parameter γ represents the weight of the inflation preference (relative to unemployment) chosen by the policymaker. Meanwhile, $k < 1$ reflects distortion that compels the government to target unemployment below equilibrium. The indifference curve representing S is presented in Fig. 10.1.

Assuming no uncertainty, $\pi^e = \pi_t$ and, therefore, $u_t = u^*$. If, in period $t - 1$, the policymaker (government) is committed to controlling inflation, remembering $u_t = u^*$, the optimal rate of inflation is $\pi_t = 0$. Therefore, the optimal *ex ante* choice is $(u^*, 0)$, namely at point O.

Nevertheless, the commitment is time inconsistent because in the subsequent period (*ex post*), when the economic agents' expectations have already been formed (in line with inflation set by the government in period t), the government

⁵The S curve is concave with a negative partial derivative to both arguments, u_t and π_t .

is tempted to push up inflation to reduce unemployment further. Ultimately, this will edge up inflation and create inflation bias, namely at point C. It is said to be a bias because with the rational expectations of economic agents, the main preference of the policymaker to reduce unemployment and also the commitment to control inflation, only higher inflation would be achieved, while unemployment would not be reduced.

10.3.2. Barro-Gordon Model

The ideas of Kydland and Prescott were more clearly interpreted by Barro and Gordon (1983) in a model which showed that through commitment (rule), the central bank could achieve a desired level of inflation consistent with public expectations, while incurring minimal losses. With discretion, however, public expectations constantly change in line with the inconsistency of the central bank's measures, thus creating inflation bias.

Barro and Gordon (1983) analyzed the time inconsistency problem in monetary policy through game theory, à la Nash Equilibrium, between the central bank and public or private sector in the economy with the following conditions:

- (1) The central bank is assumed capable of managing economic activity and orienting monetary policy toward achieving social prosperity/public welfare.
- (2) The public only acts in accordance with trusted inflation expectations.
- (3) Time inconsistency can occur because on one side the public must form their inflation expectations at the beginning of the period and maintain them until the end of the period, while, on the other hand, the central bank has full discretion to set strategy over time.

Under such conditions, the inflation target set at the beginning of the period may not be optimal by the end of the period and will create gains/incur costs for the central bank and public.

The model was formulated as follows. The central bank minimizes the social loss function (social costs):

$$L = b(U - U^*)^2 + \pi^2 \quad (11)$$

with parameter $b > 0$ and initial inflation target of zero ($\pi^* = 0$).

The relationship between unemployment (U) and inflation follows the EAPC as follows:

$$U = U^n - a(\pi - \pi^e) \text{ with } a > 0 \text{ and } U^* = k U^n \text{ where } 0 < k < 1 \quad (12)$$

The central bank can control inflation through its monetary policy, thus inflation is in line with money supply growth, namely $\pi = \mu$. In other words, there are

no problems in terms of monetary policy transmission. With that assumption, the central bank's problem is to minimize the social costs as follows:

$$Z = \{b[(1-k)U^n - a(\pi - \pi^e)]^2 + \pi^2\} \quad (13)$$

From the first-order condition, the social costs would be minimized if inflation is reduced optimally as follows:

$$\begin{aligned} \partial Z / \partial \pi &= -2ab[(1-k)U^n - a(\pi - \pi^e)] + 2\pi = 0 \\ \pi^{**} &= ab(1-k)U^n / (1 + a^2b) + a^2b \pi^e / (1 + a^2b) = \Phi(\pi^e) \end{aligned} \quad (14)$$

The solution indicates time inconsistency, namely that the initial inflation target of zero ($\pi^* = 0$) is no longer optimal by the end of the period. The presence of EAPC causes the optimal rate of inflation to be influenced by inflation expectations. In general, inflation expectations are positive ($\pi^e > 0$), therefore optimal inflation is also positive ($\pi^{**} > 0$). In fact, if inflation expectations are zero ($\pi^e = 0$), optimal inflation would still be positive ($\pi^{**} > 0$) because of the social cost factor (b) and unemployment phenomenon (k) as well as inflation deviation from the target (a), which is positive.

Why? This is because public inflation expectations influence actual inflation, which can be explained by the sources, namely:

- (a) *Surprise inflation.* Inflation will still occur even if the central bank sets the initial inflation target at zero, $\pi^* = 0$, and public expectations are also zero, $\pi^e = 0$, as long as the central bank can ignore inflation deviation from the target because of real sector dynamics. Under such conditions, actual inflation is as follows:

$$\pi^s = ab(1-k)U^n / (1 + a^2b) \quad (15)$$

Thus, the social cost of inflation is as follows:

$$\begin{aligned} L^s &= \{b[(1-k)U^n - a(ab(1-k)U^n) / (1 + a^2b)]^2 + [ab(1-k)U^n / (1 + a^2b)]^2\} \\ L^s &= b[(1-k)U^n]^2 / (1 + a^2b) \end{aligned} \quad (16)$$

- (b) *Inflation bias.* Although the public form rational expectations (congruent with the rational expectations hypothesis), thereby knowing the central bank loss function and Phillip's curve behavior (game theory à la Stackelberg), the initial inflation target, $\pi^* = 0$ would not be credible. Under such conditions, inflation expectations would be the same as the central bank's target, $\pi^* = \pi^e$. Therefore, actual inflation would become:

$$\pi^{\text{REH}} = ab(1-k)U^n \quad (17)$$

And the social cost:

$$L^{\text{REH}} = b[(1-k)U^n]^2 + ab(1-k)U^n = b(1+a^2b) [(1-k)U^n]^2 \quad (18)$$

- (c) *Inflation rule.* Barro and Gordon suggested that with rational expectations and the behavior/game above, the social cost could be reduced by giving the central bank a rule that inflation = 0 and money supply growth = μ . Under such conditions:

$$\pi^{\text{RULE}} = 0 \text{ and } L^{\text{RULE}} = b[(1-k)U^n]^2 \quad (19)$$

The analysis above indicates that the smallest social cost occurs when there is surprise inflation and largest when there is inflation bias, while the inflation rule is somewhere between the two or:

$$L^S < L^{\text{RULE}} < L^{\text{REH}} \quad (20)$$

Therefore, in the context of game theory, the Barro-Gordon model tends to produce the prisoner's dilemma, meaning that the optimal strategy for both agents (central bank and public) would be detrimental to both agents.

The discussion above shows that while there is deviation between the inflation target and actual inflation, optimal inflation will not be achieved, which incurs a social cost. Consequently, the central bank is required to orient policy to achieve the target and the public form inflation expectations in line with the target set by the central bank, thus $\pi^* = \pi^e = \pi^{**}$. This is where the issue of monetary policy credibility becomes important.

To explain the benefits of monetary policy credibility, the Barro-Gordon model can be reanalyzed as follows:

- (a) If monetary policy is not credible, the central bank is forced to assume public inflation expectations as an (exogenous) explanatory variable. Therefore, as in the previous analysis, the social cost will be minimized if:

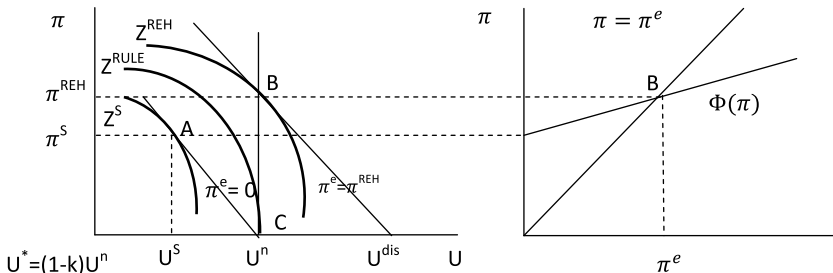


Fig. 10.2: Policy Strategy and the Social Cost.
Source: Bofinger (2000).

$$\pi^{**} = ab(1-k)U^n / (1+a^2b) + a^2b \pi^e / (1+a^2b) = \Phi(\pi^e) \quad (21)$$

In this case, the central bank would expect optimal inflation policy to be influenced by inflation expectations and the level of unemployment, hence the inflation target would change over time. The level of unemployment at optimal inflation is as follows:

$$\begin{aligned} U^{**} &= U^n - a[\{ab(1-k)U^n / (1+a^2b) + a^2b \pi^e / (1+a^2b)\} - \pi^e] \\ &= [1 - a^2b(1-k) / (1+a^2b)]U^n - a[a^2b / (1+a^2b) - 1]\pi^e \end{aligned} \quad (22)$$

$$\text{So } U^{**} = [(1+a^2bk) / (1+a^2b)]U^n + [a / (1+a^2b)]\pi^e$$

*****Therefore, the actual level of unemployment depends on the target (value) of desired unemployment, k , and inflation expectations. The natural rate of unemployment under economic equilibrium will be achieved if the target $k = 1$ and inflation expectations $\pi^e = 0$.

- (b) On the other hand, if monetary policy is credible, thus $\pi = \pi^* = \pi^e$, then the social cost is:

$$L = \{b[(1-k)U^n]^2 + \pi^2\} \quad (23)$$

From the first-order condition:

$$\partial L / \partial \pi : 2\pi = 0 \quad (24)$$

Therefore, if monetary policy is credible, optimal inflation is $\pi^{**} = 0$. Such conditions also guarantee achievement of the natural rate of unemployment.

10.3.3. *Credibility Problem Analysis with Supply Shocks*

Sources of the policy credibility problem are very diverse and complex. From a theoretical perspective, that rationale highlights the role of changes in the preferences of the policymakers to stimulate output or reduce unemployment (parameters b and k) as well as inflation deviation due to the need to stimulate output or reduce unemployment (parameter a).

What follows is an analysis of how the credibility problem could be exacerbated by shocks in the economy, for instance supply-side shocks. This is important to monitor, primarily by considering the economic characteristics of developing countries, where supply shocks are comparatively dominant in terms of influencing monetary policy effectiveness.

Therefore, the Barro-Gordon model was developed further, namely by including supply shocks. This implies tighter requirements for the central bank when formulating monetary policy to garner credibility. To analyze monetary policy

credibility under conditions of supply shocks, the Phillip's curve is assumed to behave as follows:

$$U = U^n - a(\pi - \pi^e) + \varepsilon \quad (25)$$

where ε = supply shock.

Therefore, the problem of minimizing the social cost by the central bank becomes:

$$\begin{aligned} L &= [b(U - U^*)^2 + \pi^2] = \{b[U^n - a(\pi - \pi^e) + \varepsilon - kU^n]^2 + \pi^2\} \\ L &= \{b[(1-k)U^n - a(\pi - \pi^e) + \varepsilon]^2 + \pi^2\} \end{aligned} \quad (26)$$

Consequently, three scenarios can be analyzed as follows: According to the first scenario, the central bank cannot guarantee that actual inflation will be zero but can be committed to maintaining average inflation at zero ($\pi^R = 0$). The public trusts the central bank and forms expectations based on the average inflation, thus $\pi^e = \pi^R = 0$. Therefore, the social cost becomes:

$$L = [b((1-k)U^n - a\pi + \varepsilon)^2 + \pi^2] \quad (27)$$

From the first-order condition:

$$\begin{aligned} \partial L / \partial \pi &= -2ab((1-k)U^n - a\pi + \varepsilon) + 2\pi = 0 \\ \pi &= ab((1-k)U^n + \varepsilon) / (1 + a^2b) \end{aligned} \quad (28)$$

$$\text{And average inflation is: } E(\pi) = ab(1-k)U^n / (1 + a^2b) \quad (29)$$

According to this scenario, therefore, to ensure average inflation is equal to zero ($E(\pi) = 0$), the central bank would need to not only overcome the supply shocks but also orient monetary policy to achieve full employment output, or $k = 1$. Monetary policy credibility would produce an optimal rate of inflation $\pi^{**} = 0$. Therefore, although credibility could be achieved at any level of inflation, optimal inflation policy (inflation = zero) would create full credibility. Under such conditions, the natural rate of unemployment could also be achieved.

According to the second scenario, the central bank guarantees that actual (not average) inflation would be equal to zero and the public trusts the central bank, thus $\pi^e = \pi = 0$. Therefore, the social cost becomes:

$$L = [b((1-k)U^n + \varepsilon)^2 + \pi^2] \quad (30)$$

And the inflation solution is $\pi = 0$

If the central bank could achieve actual inflation of 0, the policy would be considered credible although difficult to implement because the contained supply shocks are difficult to predict.

Per the third scenario, the central bank treats public inflation expectations (π^e) as exogenous and the public act rationally. The social cost therefore becomes:

$$L = [b((1-k)U^n - a(\pi - \pi^e) + \varepsilon)^2 + \pi^2] \quad (31)$$

From the first-order condition:

$$\partial L / \partial \pi = -2ab((1-k)U^n - a(\pi - \pi^e) + \varepsilon) + 2\pi = 0 \quad (31)$$

Actual inflation becomes:

$$\pi = ab((1-k)U^n + a\pi^e + \varepsilon) / (1 + a^2b) \quad (32)$$

And average inflation:

$$E(\pi) = ab((1-k)U^n + a\pi^e) / (1 + a^2b) \quad (33)$$

According to this scenario, it is more difficult for the central bank to achieve monetary policy credibility because the central bank must meet several requirements: (1) achieve the inflation target and earn public trust, thus the inflation expectations are equal to zero ($\pi^e = 0$); (2) overcome supply shocks; and (3) orient monetary policy toward achieving full employment output, or $k = 1$.

From the three scenarios, the second was considered the best. Therefore, monetary policy would gain credibility if the inflation target was achieved along with public trust in the formation of inflation expectations. Nevertheless, the impact on output and unemployment would also have to be analyzed.

10.4. Empirical Studies: Measures of Credibility and Time Inconsistency

As mentioned previously, numerous studies have been conducted regarding the credibility problem and time inconsistency but most are hypothetical, namely with a fundamental understanding that the time inconsistency problem would create inflation bias and then reduce the effectiveness of monetary policy. Very few empirical studies deal with measuring the extent of the costs incurred due to time inconsistency. In general, the empirical studies deal with the influence of institutional aspects, for instance independence and transparency, in terms of the impact on monetary policy effectiveness. Several relevant empirical studies dealing

with measures of credibility and time inconsistency, using various approaches, are detailed as follows.

10.4.1. Parameterization Approach: Credibility Index

Cobham, Papadopoulos, and Zis (2001) analyzed the credibility of monetary policy implementation in the UK for the period from 1994 to 1997 using an *ex ante* and *ex post* credibility index. The *ex ante* credibility index was compiled based on the frequency of agreement/disagreement between the Minister of Finance as the decisionmaker and the Bank of England (BoE) as the provider of recommendations concerning changes to the interest rate. Meanwhile, the *ex post* credibility index was calculated based on the yield differential between the UK and Germany (or the US). Arguing that political intervention to manipulate central bank preferences would reduce monetary policy credibility, they hypothesized that there is a correlation between both aspects.

For example, a lack of agreement from the Minister of Finance regarding the recommendations put forward by the BoE (published with a certain lag) could be interpreted by the financial markets as a difference in preferences between the government and central bank (mandated with maintaining price stability). This could be further interpreted as evidence that the political interests of government to realize a separate goal would undermine monetary policy credibility, thereby raising the long-term interest rate. Using the long-term interest rate relates to the understanding that the long-term interest rate contains a risk premium, which reflects the possibility that the timing and magnitude of interest rate change is influenced by political considerations. In that context, long-term interest rate data is taken from the yield data of 10-year government debt securities.

Meanwhile, the classification of agreement/disagreement used to compile the *ex ante* credibility index refers to the five categories obtained from the minutes of the meeting, namely a firm vote to raise the interest rate, propensity to raise the interest rate, impartiality to raise/lower the interest rate, propensity to lower the interest rate, and a firm vote to lower the interest rate. Statistically, the index with an interval of [0,1] has a number of characteristics. *First*, this index is sufficiently persistent, where the value is influenced by the previous value (smoothing parameter of 0.75) and adjusted incrementally to the latest information regarding new consensus. *Second*, the index behaves asymptotically toward the extremes (between 0 and 1), implying that for the next agreement/disagreement to occur, the progressive increase/decrease in the index will become smaller.

The next step is to assess the equations of long-term correlation of the two credibility indexes. The assessment evidenced a negative correlation between the two variables, namely that monetary policy credibility in the UK is symmetrically linked to the agreement/disagreement between the Minister of Finance and BoE. Those findings imply that the costs incurred due to political intervention on monetary policy implementation are significant. It was concluded from the simulations that if the Minister of Finance agrees with the recommendations put forward by the BoE, the average long-term interest rate will be 1.5% lower than

the average actual rate. In other words, if the long-term interest rate in Germany was 5.87% (April 1997), without political intervention, the long-term interest rate in UK would be 6.11%, not the 7.61% recorded.

Interesting results are found relating to an alternative assessment if different long-term interest rates in the United Kingdom and United States are used. The assessment found that there was not a symmetrical correlation between different long-term interest rates in the United Kingdom and United States with the credibility index, which reflects differences in terms of monetary policy implementation or other specific factors that influence yields in the United States and Germany. In this case, they found that United States yields were more fluctuative than yields in Germany. Ultimately, the differences between the United States and Germany were easier to understand in a more strategic context, where the United States monetary policy regime was considered to lack a clear nominal anchor, more commonly referred to as a “just do it” regime.

10.4.2. Qualitative Approach (Narrative)

Researching the relevance of time inconsistency theory in terms of setting monetary policy in the United States, Chappel and McGregor (2004) applied a narrative approach, namely by using historical evidence in the form of recorded conversations at the Federal Open Market Committee (FOMC) contained in the Memorandum of Discussion to evaluate the verbal arguments used by policymakers to support their desired policy direction. Empirically, the analysis focused on the period from 1970 to 1978 (when the Fed was under the leadership of Arthur Burns), a period when strong inflationary pressures blighted the United States. The main advantage of the narrative approach is its original ability and fairness to explore the rationality and motives of the policymakers through their arguments delivered to support a specific policy direction. Nevertheless, the approach does not allow the testing of a hypothesis against other alternative hypotheses, while the arguments observed were chosen beforehand in the context of the analysis.

The analysis based on the Memorandums of Discussion and various other transcripts obtained from official documentation (Ford Library) explored the journeys of the policies containing time inconsistency during the 1970s, which could be grouped into four aspects as follows: (1) public and political preferences concerning economic performance; (2) the short and long-term trade-offs of the Phillips curve; (3) the policymaking characteristics of the FOMC over time; and (4) the role of expectations under equilibrium conditions.

Broadly speaking, based on those four aspects, the narrative analysis produced the following conclusions:

- (1) In general, the FOMC under Arthur Burns very rarely achieved politically acceptable levels of output and unemployment. Technically, in the standard Barro-Gordon model, inflation performed beyond the desired level ($b > 0$), while political pressures forced the Fed to constantly strive toward a level of unemployment below the natural rate ($k < 1$).

- (2) The FOMC adhered to the Phillips curve behavior assumed in the time inconsistency theory. In the short term, the sacrifice ratio was relatively high but in the long term, the inflation expectations formed contributed to high inflation and unemployment. In addition, from the FOMC's perspective, the rising cost of controlling inflation, triggered by expectations, was considered exogenous at each meeting. Consequently, although inflation was already high, the rational measures taken by the policymakers tended to perpetuate high inflation.
- (3) During the 1970s, FOMC members faced problems in terms of determining the policy focus for the subsequent period as well as considering the long-term consequences of that policy. One member complained about the briefness of the time allocated to formulate and deliver alternative policies to the FOMC, while also believing a longer period was required to evaluate the future inflationary impact of the policies proposed. Furthermore, the FOMC desired contractionary monetary policy with the consequence of a possible recession. Nevertheless, pressures in the form of public and political opinion blocked that decision. Such conditions demonstrated how the FOMC was forced into policies with short-term goals, despite knowing there would be long-term consequences.
- (4) Although FOMC members tended to regard public expectations as inertia, they also felt that the financial markets often reacted excessively and irrationally in response to changes in policy. Under such circumstances, the FOMC tended to be too prudent in response to market behaviors when interpreting a policy and tried to avoid excessive and irrational market reactions. Consequently, the FOMC realized that the benefits of the monetary stimuli issued would be more effective if applied without a corresponding increase in public expectations.

From the narrative, the problem of time inconsistency was ultimately concluded to play an important role in terms of perpetuating inflation persistence in the United States during the 1970s. In this case, the Fed admitted that the Phillips curve trade-off and political pressures had made it more difficult for the Fed to introduce disinflationary policies in each period. On the other hand, however, such conditions also exacerbated the already excessive expansionary monetary policy due to an extremely short policy planning horizon at each FOMC meeting.

10.4.3. Quantitative Modeling Approach

Analyzing the phenomenon of time inconsistency in the United States, Surico (2003) assessed the magnitude of inflation bias in monetary policy implementation for the period from 1960 to 2002 using a hypothetical model, representing the reduced form of the policy response (policy rule). Surico developed a strategy to identify the relevant asymmetric parameters in the objective function of the central bank, where policymakers were more concerned about output contraction rather than expansion, and then interpreted that into a measure of

time inconsistency. In addition, by comparing solutions with commitment and solutions using discretion, average inflation could be decomposed in the form of a target and bias. This approach provided an important new contribution in the literature about optimal monetary policy because it could measure the average inflation bias that appeared from the asymmetrical preferences of the central bank on output stabilization.

The central bank loss function in exponential linear form is specified as follows:

$$L_t = \frac{1}{2} (\pi_t - \pi^*)^2 + \lambda[(\exp(\gamma y_t) - \gamma y_t - 1) / \gamma^2] \quad (34)$$

Meanwhile, the feedback rule of the empirical model, solved by optimizing the central bank objective function is as follows:

$$\pi_t = \pi^* + \alpha y_t + \beta y_t^2 + v_t \quad (35)$$

That equation has a linear parameter, $\alpha = -\lambda\theta$ and $\beta = \lambda\theta / 2$

With the error terms defined as follows:

$$v_t = -\{\alpha(y_t - E_{t-1}y_t) + \beta[y_t^2 - E_{t-1}(y_t^2)] + e_t\}$$

where y is the output gap, λ is the relative weight of the central bank preference to output stability, θ reflects the effect of inflation deviation (on expectations) in the output formation of the Phillips curve, and γ refers to the characteristics of the asymmetric central bank preferences. Meanwhile, rational expectations are formed in period $t - 1$. The complexity of the parameters in the equation represents the preferences of the policymaker and explains the structure of the economy. By solving the reduced form, the asymmetric preferences of the central bank can be identified further, $\gamma = 2\beta/\alpha$ and average inflation bias, $\beta\sigma_y^2$.

If the costs that are incurred are larger during output contraction than output expansion, the model predicts $\gamma < 0$, $\alpha < 0$ (because γ and θ are positive) and $\beta > 0$. By including the behavior in the Phillips curve, the model implies that the central bank has an incentive for surprise inflation to negate the possibility of declining economic activity. Therefore, the asymmetric preference for output stability would precipitate prudence in terms of expansion. This is congruent with the model behavior that predicted a positive correlation between average inflation and variations in the output gap (σ_y^2).

Observations were also performed covering two different regimes, namely before and after the leadership of Paul Volker (1979 breakpoint) as well as during the period of leadership under Alan Greenspan (after 1987). Using the Generalized Method of Movement (GMM), the results concluded there were significant policy (regime) changes during 40 years of monetary policy implementation in the United States, as measured by changes in the monetary policy preferences of the Federal Reserve. Specifically, the results showed that in addition to a reduction in average inflation target from 3.42% to 1.96%, inflation bias, which was estimated at around 1.01% prior to 1979, was minimal (approaching zero) in the

final two decades of the observation period. The rationality of the results can be linked to the magnitude and asymmetry of the monetary policy preference toward output stabilization found in the period of Paul Volker's leadership and before, but not in the period thereafter. Despite other affecting factors, including better policy formulation and the minimal impact of supply shocks, the analysis successfully provided a new notion in the empirical evidence and quantitative measures of US policy behavior.

10.5. Assessing Monetary Policy Credibility in Indonesia

A change to the mandate, policies and institutional arrangements of Bank Indonesia after the passing of Act No. 23 of 1999 eventually oriented monetary policy in Indonesia toward implementation of the inflation targeting framework (ITF). Bank Indonesia reforms consolidated the institutional dimension (governance) in terms of central bank practices and simultaneously represented an integral part of macroeconomic recovery policy to escape the clutches of crisis.

10.5.1. Policy Credibility in Indonesia during ITF Implementation

Per the ITF regime, forward-looking monetary policy to future inflation implies that future inflation must be controlled in line with the existing target. To support policy formulation, the central bank must be able to capture or take into account expectations as well as accurate and reliable information concerning future inflation projections to set the desired monetary policy response, be it neutral, tighter or looser. This surely requires high policy credibility at the central bank to provide avowed commitment to achieving low and stable inflation. If the central bank is credible, the inflation expectations of economic agents could be oriented toward the inflation target set by the central bank. As explained previously, theoretically, if monetary policy is fully credible, inflation expectations and actual inflation will converge on the inflation target.⁶

Observations of the Indonesian case have shown that during the past decade, inflation has tracked a downward trend toward the era of single-digit inflation. In this case, congruent with increasing inflation control policy credibility by Bank Indonesia working in conjunction with the Government during the ITF era since 2005, core inflation has declined significantly from an average of 8.1% before

⁶In practical terms, however, policy credibility can be observed in terms of target achievement and policy implementation. Concerning policy target achievement, credibility can be assessed based on whether the central bank can attain the inflation target. Meanwhile, regarding the policy implementation process, policy credibility can be evaluated based on the policy response, whether implemented consistently or not. Theoretically, both aspects are closely related, implying that a well-planned policy implementation process will support achievement of the desired policy target. In practice, however, the central bank may implement sound and consistent policy but, due to influences beyond the control of the central bank, the inflation target cannot be achieved optimally.

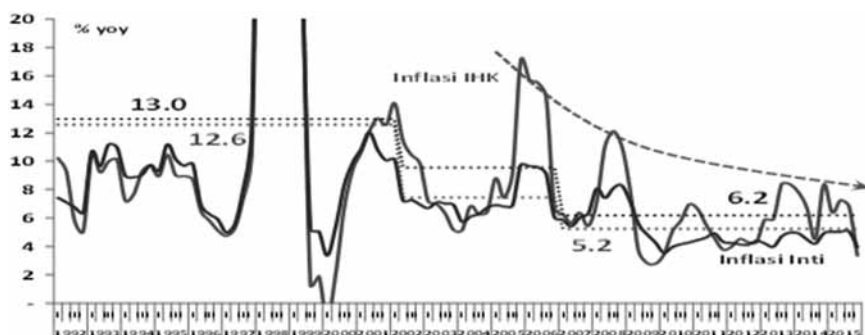


Fig. 10.3: Core and Headline Inflation Trends.

ITF implementation to an average of 5.2% after ITF implementation. Likewise, headline CPI inflation has also experienced significant disinflation from an average of 8.7% to 6.2%. It is important to note that during a few episodes, headline inflation exceeded the target set by the Government, after consultation with Bank Indonesia, due to extreme inflationary pressures stemming from administered prices (AP) and volatile foods (VF). Moving forward, however, the government's energy reforms have bolstered commitment to controlling inflation on the supply side, therefore, inflationary pressures from AP and VF are predicted to remain minimal.

Such developments are also in line with the results of empirical studies showing that prior to ITF implementation, monetary policy credibility was relatively low in Indonesia and this was one of the determinants of high inflation persistence (Harmanta, 2009). Nonetheless, congruent with ITF implementation, monetary policy credibility improved, reflected by a Kalman gain⁷ from around 0.2 in the period before ITF implementation (July 2005) to approximately 0.4 after ITF implementation.

It could be argued that if monetary policy credibility was still low, similar to conditions before ITF implementation, inflation would be influenced by the backward-looking behavior of economic agents, thus creating higher inflation persistence and delaying convergence with the central bank's inflation target. This is because monetary policy that is not yet credible prolongs the learning process of economic agents in the context of the inflation target. In contrast, when monetary policy credibility improves, such as after ITF implementation, inflation

⁷Technically, the Kalman gain captures the forecasts of economic agents in terms of projecting the inflation target of the monetary authority in the Taylor rule equation through a derivation of the Kalman filter. The Kalman gain reflects the learning process of economic agents towards the inflation target of the monetary authority, which can be used as a measure of the degree of monetary policy credibility, where a faster learning process implies a greater degree of monetary policy credibility. The value of the Kalman gain is between 0 and 1, where a value closer to 1 indicates a greater degree of monetary policy credibility.

is then influenced more by forward-looking expectations, which lowers inflation persistence and anchors inflation expectations to the monetary authority's target that is also forward looking.⁸

The dimensions of monetary policy credibility can be observed from the implementation of inflation (disinflation) control strategy. Theoretically, ITF aims to achieve low and stable inflation.⁹ Empirically, based on the experience of several ITF countries, there are a number of strategies that could be employed to achieve low and stable inflation. In general, when ITF was first implemented, advanced countries had typically already achieved low (single-digit) inflation, thereby facilitating an aggressive long-term disinflation strategy to around 2.0%–3.0%. In contrast, developing countries were typically experiencing double-digit inflation when ITF was first implemented, thus a more gradual disinflation strategy was implemented with a short- and medium-term target and the pace of disinflation was set at around 0.8% per annum prior to convergence with the long-term inflation target of 2.0%–3.0% (Roger & Stone, 2005).¹⁰

Observations of the Indonesian case have shown that the disinflation process pursued by Bank Indonesia during the ITF era, including the transition phase and after official ITF implementation, was applied gradually. The rate of disinflation stood at around –0.5% per annum, slightly lower than the average disinflation rate recorded in developing countries at –0.7% per annum (Roger & Stone, 2005). This is in line with monetary policy in Indonesia, which is not fully credible, thus the learning process of economic agents in terms of reducing the inflation target to low and stable inflation was slightly more protracted. In addition, the

⁸This is congruent with numerous empirical observations showing that the formation of natural inflation expectations oriented towards the inflation target as the main monetary policy target in the ITF era would facilitate the learning process of economic agents compared to conditions before ITF application, when multiple targets were possible. That process would create more forward-looking inflation expectations among economic agents after ITF implementation, which is in line with the findings of Juhro (2004), Yunuarti (2007), and Alamsyah (2008). Furthermore, that research is also consistent with Siregar and Goo (2008), who showed that inflation inertia in Indonesia declined, for tradeable and non-tradeable goods, after ITF implementation.

⁹Experience from several ITF countries showed that the inflation target has a number of traits: (1) sufficiently low; (2) small fluctuations; (3) challenging for the central bank to achieve; and (4) can be achieved with minimal output loss.

¹⁰Several studies have shown that the disinflation strategy is explicitly subject to central bank monetary policy credibility. Advanced countries, where monetary policy is deemed more credible, tend to apply a more aggressive disinflation strategy because monetary policy can anchor the inflation expectations of economic agents and actual inflation with greater rapidity towards the inflation target, thus the interest rate response will be optimal and avoid any large output loss. Developing countries, however, where monetary policy is considered less credible, tend to apply a more gradual disinflation strategy because monetary policy is unable to steer inflation expectations and actual inflation as rapidly towards the inflation target, hence the interest rate response is not optimal and could trigger a large output loss.

gradual disinflation process implemented in Indonesia was also due to previous experience of double-digit inflation as well as the moderate degree of inflation persistence, while not fully forward looking or anchored to the inflation target.

Gradualism in terms of the disinflation process in Indonesia was fundamental because of imperfect credibility, therefore overly aggressive monetary policy oriented toward disinflation within a brief period of time would have led to an unacceptably high sacrifice ratio. In the context of a small open economy, the impact of rapid disinflation against a backdrop of imperfect credibility by hiking the interest rate would spur excessive appreciatory pressures on the exchange rate, thus exacerbating the trade-off between inflation and output.¹¹

With the gradual disinflation strategy adopted in Indonesia, achieving the long-term target of low and stable inflation was expected to require a prolonged period. Nevertheless, this was also affected by the process to build policy credibility itself and the dynamic public learning process. In relation, the empirical study conducted by Cukierman (2006) showed that a characteristic of inflation stabilization from double digits toward the medium- and long-term inflation target was gradual disinflation. Experience in several countries, including Chile, has also shown that disinflation toward low and stable inflation under the ITF requires a prolonged period, namely around 36 quarters, or 9 years. In fact, in addition to Indonesia, there remain several ITF developing countries that are yet to complete the disinflation process, including Colombia – since 1999, Rumania – since 2005, and Turkey – since 2006.

10.5.2. Empirical Studies of Time Inconsistency in Indonesia

Harmanta (2009) reported that high inflation persistence in Indonesia is caused by imperfect monetary policy credibility. Furthermore, research by Revenna (2005) also substantiated the previous claims based on a survey of 82 countries that placed monetary policy in Indonesia in the “low credibility” category because the inflation target had not been achieved. The issue of monetary policy credibility in Indonesia relates closely to the time inconsistency problem, (Goeltom, 2005). Time-inconsistent policy has the potential to undermine monetary policy credibility, thus the economic agents will form higher inflation expectations than the target announced by the authorities. If this is the case, does the credibility problem in Indonesia indicate time inconsistency in terms of Bank Indonesia’s monetary policy? Goeltom (2005) stated that from 1990 to 2003, monetary policy in Indonesia was subject to the problem of time inconsistency, as reflected in the

¹¹Due to imperfect credibility, a gradual interest rate decline when shocks occur in terms of lowering the inflation target and actual inflation, as reflected in a tight bias monetary policy stance, could lead to a larger output loss. Tight bias monetary policy would influence the consumption habits of economic agents and, hence, weaken aggregate demand, which would feed through to the production side and output would therefore not increase. Such conditions could create a larger output loss, as reflected in the sacrifice ratio.

suboptimal monetary policy, which manifested in an overly loose policy stance during an economic upswing and excessively tight monetary policy when the economy flagged.

Studies on the institutional arrangements of the central bank are a nascent topic today and, therefore, not enough research has been done, especially for the Indonesian case. Studies on time-inconsistent monetary policy in Indonesia remain scarce. One study, conducted by Budiyantri (2009) revealed the problem of time-inconsistent monetary policy in Indonesia before and after the Asian Financial Crisis of 1997. Budiyantri used the Barro-Gordon standard linear quadratic model to explain the problem of time-inconsistent monetary policy in Indonesia. Fundamentally, the model assumed a symmetrical central bank preference to the output gap, implying that the central bank was indifferent to a positive or negative output gap.

Nevertheless, the assumption of symmetrical preferences received criticism from academics and monetary practitioners on a theoretical and practical level (Blinder, 1998; McCallum, 1997). Empirically too, a large body of research doubted the assumptions of the standard linear quadratic model to explain the time inconsistency problem (Cukierman, 2000; Surico, 2003). Cukierman, for instance, questioned whether central banks really had the same preference toward negative and positive output deviation from its potential? Observations showed that the central bank and the public tend to prefer a negative output gap during an economic downturn rather than a positive output gap during an economic upswing. Therefore, such asymmetric preferences have become an important issue and interesting study topic, particularly in terms of time-inconsistent monetary policy by the central bank.

The study by Rahmahdian and Warjiyo (2013) was the first to measure asymmetric preferences in terms of time-inconsistent monetary policy in Indonesia. They found at least three strategic issues in their research. *First*, was there an asymmetric preference parameter to the magnitude of the output gap, thus indicating time-inconsistent monetary policy in Indonesia before and after Bank Indonesia's independence? If yes, was there a difference in the magnitude of the asymmetric preference, thus indicating different degrees of time inconsistency before and after Bank Indonesia's independence? *Second*, what were the implications of time-inconsistent monetary policy or policy with different degrees of time inconsistency before and after Bank Indonesia's independence in terms of influencing the output gap on setting the inflation target in Indonesia? *Third*, how did inflation perform in Indonesia before and after Bank Indonesia's independence in the context of time-(in)consistent monetary policy or with differing degrees of time inconsistency?

The analysis period was separated into two subsamples, namely the period prior to Bank Indonesia's independence (1990:1 to 1999:4) and the period after independence (2000:1 to 2009:4). This was congruent with a fundamental change in the institutional arrangements and application of monetary policy in Indonesia in 2000, marked by a change in Bank Indonesia's status from dependent to independent pursuant to Act No. 23 of 1999, which explicitly stipulated that monetary policy would only focus on the overriding goal of price stability.

The sample was also separated in reference to Rogoff (1985), who stated that central bank independence could overcome the time inconsistency problem. Likewise, the degree of time inconsistency was also suspected to differ in the periods before and after Bank Indonesia's independence.

The research by Rahmahdian and Warjiyo (2013) revealed several salient conclusions. *First*, the problem of time inconsistency was detected in Bank Indonesia's monetary policy prior to independence, with an asymmetric preference to the output gap, contrasting the time-consistent monetary policy instituted after Bank Indonesia's independence, with a symmetrical preference to the output gap. *Second*, time-inconsistent monetary policy prior to independence triggered intense output gap pressures on inflation, while time-consistent monetary policy after independence eliminated the influence of the output gap on inflation. *Third*, the implementation of consistent ITF after independence has successfully realized lower inflation, although the disinflation process has been prolonged and not quite as expected.

The empirical findings of an asymmetric preference toward the output gap and its impact on time-inconsistent monetary policy before independence was reputedly due to government control over Bank Indonesia at that time. Pursuant to Act No. 13 of 1968, Bank Indonesia was mandated with multiple objectives, namely: *first*, to control, create and maintain rupiah stability and *second*, to stimulate production and development as well as expand job opportunities to ameliorate the public standard of living. That double mandate led to an asymmetric preference toward the output gap because, in addition to maintaining price stability, Bank Indonesia was also an agent of development, obligated to create job opportunities. These findings corroborated the previous research conducted by Budiyaniti (2009), who found that during the period before the crisis (1990–1997), Bank Indonesia faced the time inconsistency problem in the short and long term. Furthermore, the research was also consistent with Goeltom (2005), who stated that Bank Indonesia's monetary policy from 1990 to 2003 still faced the problem of time inconsistency, which manifested in too tight and sometimes too loose monetary policy.

Regarding the subsample after Bank Indonesia's independence, the time inconsistency problem was no longer detected. Bank Indonesia's preference to the output gap had become symmetrical, positive or negative, thus reflecting consistent monetary policy and commitment to achieve low inflation by reducing the element of discretion in response to current economic dynamics (boom or recession). These findings were ostensibly due to the single mandate of rupiah stability and independent status of Bank Indonesia in accordance with Act No. 23 of 1999. Furthermore, monetary policy consistency, coupled with a symmetrical preference toward the output gap, was also observed to improve ITF performance. More consistent monetary policy led to more credible monetary policy in the eyes of economic agents. The empirical evidence for monetary policy consistency after independence was also accompanied by increased credibility at Bank Indonesia. Harmanta (2009) reported that monetary policy credibility increased at Bank Indonesia after ITF implementation despite imperfect credibility, which is understandable considering full ITF implementation is relatively new.

The aforementioned three conclusions have salient implications for monetary policy in Indonesia. *First*, Bank Indonesia is required to constantly improve monetary policy consistency. The effectiveness of achieving the final target of monetary policy with the single mandate of price stability depends on how committed Bank Indonesia is to avoiding the inconsistency problem in the near term in order to realize low and stable inflation, for instance by stimulating stronger economic growth. It would be better to implement more consistent monetary policy based on clear rules, while reducing the element of discretion. Monetary policy consistent with ITF has been proven to lower inflation despite the current delays. Therefore, to restore public confidence in the reputation of Bank Indonesia, monetary policy consistency to achieve stability must constantly be improved.

The *second* implication is the need to increase coordination. Considering that not all components of inflation are influenced by monetary policy, in particular AP and VF, close coordination between monetary policy and government policy, including the central government and regional administrations, is required. Existing coordination between the Government and Bank Indonesia, especially through the National Inflation Task Force (TPI) and Regional Inflation Task Forces (TPID), must be improved to achieve price stability as a prerequisite for ameliorating public welfare. In addition to minimizing the inflationary pressures originating from rising AP and VF, policy coordination is also critical to strengthen synergy in terms of general economic management.

The *third* implication is the need to increase communication. Anchoring inflation expectations is an integral element of the ITF-based monetary policy framework, considering how far inflation expectations determine actual inflation. Public inflation expectations are still sometimes backward looking; therefore, more transparent monetary policy is required to reduce the asymmetric information between Bank Indonesia and the economic agents. Bank Indonesia must increase monetary policy communication in order to orient public inflation expectations toward the future (forward looking) pursuant to the requirements of ITF. The main objective of strengthening the communication strategy is to help gradually lower and anchor public inflation expectations to the inflation target.

In the Indonesian case, congruent with high inflation persistence and uncertainty, the effectiveness of achieving the final target of monetary policy with the overriding goal of price stability will depend on Bank Indonesia's commitment and credibility in terms of achieving low and stable inflation within a given horizon. In addition, monetary policy is better when instituted consistently and by reducing the element of discretion. This is linked to the large "uncertainty cost" of monetary policy implementation. Therefore, the plan to implement ITF-based monetary policy should be seen as a means to improve central bank credibility in order to overcome persistent inflationary bias and reduce the cost of controlling inflation.

Last but not least, in terms of ITF implementation, the institutional framework must be strengthened, primarily through policy coordination with the government, while aligning the spirit of independence supported by sound accountability. In addition, increased policy transparency requires the support of an effective communication strategy for all societal strata and relevant stakeholders. In this

case, the moral of the story is that macroeconomic stability should not be taken for granted but should be seen as the fruit of prudent and measured economic policy measures. An in-depth description of other institutional aspects, relating to independence, accountability and transparency as well as the monetary policy communication strategy is presented in Chapters 11 and 12.

10.6. Concluding Remarks

This chapter has discussed the salient institutional aspects of the central bank, namely policy credibility and consistency. A credible central bank can form public expectations that the policies instituted will always be oriented toward achieving the final target and that the final target can be achieved. If credibility has taken a prolong period to attain and, therefore, the central bank's reputation has already been built, monetary policy will be more effective, whether the response is through interest rates or controlling money supply. In fact, a reputable central bank and thus, strong public expectations, would be consistent with monetary policy and achievement of the final target, which could reduce and even replace the interest rate and monetary policy response.

The capacity of the central bank to build policy credibility will depend on efforts to overcome several issues, namely time-inconsistent or sector-inconsistent policy as well as asymmetric information among economic agents, economic shocks, and even political uncertainty. Therefore, to build credibility, the central bank is recommended to base the policy response on a specific rule rather than discretion, build reputation through consistent policies and strengthen political and public support. This is no mean feat because the internal institutional arrangements of the central bank itself must be strengthened, including professionalism, governance, and strong leadership to ensure avowed commitment to the mandate in accordance with prevailing regulations. Furthermore, building public perception through transparency and a clear communication strategy is also crucial.

Chapter 11

Central Bank Independence and Accountability

11.1. Introduction

Chapter 10 showed that the introduction of credibility through commitment and policy consistency will have a positive economic effect by steering the formation of public expectations. Based on public expectations, the central bank will be better equipped to boost policy effectiveness in terms of achieving price stability. Furthermore, it has been suggested that one effort to overcome the problem of policy credibility is by providing a clear mandate to conservative central bankers accompanied by full authority (independence) of central banking. In general, providing a clear mandate and independence from the principal (public) to the agent (central bank) tends to focus on the achievement of low and stable inflation. In this case, with a mandate clearly stipulated in law, the central bank is expected to enjoy independent monetary policy implementation to achieve the mandated goal. It is important to note, however, that in line with the delegation of authority, a mechanism is also ultimately required to guarantee monetary policy implementation is consistent with public expectations, which is achieved in the form of policy accountability and transparency by the central bank.

Therefore, improving policy credibility at the central bank fundamentally relates to institutional aspects, such as policy independence and accountability as well as transparency and communication. This chapter discusses aspects of central bank independence and accountability in six sections. After this introduction, the second and third sections outline the conceptual dimensions and present a literature review as well as the theoretical models linked to the need for policy independence and accountability. The fourth section will highlight several critical issues relating to the institutional aspects, including the influence of independence on economic performance and the challenges of policy independence during crisis conditions. The fifth section provides a snapshot of policy independence and accountability in Indonesia, followed by the concluding remarks, which confirm the importance of policy credibility and growing need to interpret the important role of central bank independence when confronting the complex challenges faced by the central bank.

11.2. Conceptual and Empirical Dimensions of Independence and Accountability

11.2.1. Central Bank Independence

In general, central bank independence implies that the central bank has the freedom to determine and implement monetary policy in pursuance of prevailing regulations without political considerations or interference from a third party. In fact, the issue of independence is as old as central banking itself. David Ricardo (1871) advocated central bank independence and suggested that the central bank should not finance the government's budget deficit. Considering that central bank independence is an integral part of achieving the overriding purpose of the central bank, central bank independence is often linked to institutional development and performance. Furthermore, the independence of a central bank is often tied to its ability to achieve the final target, low inflation for example. Nevertheless, the influence of independence of central bank, good and bad, remains a hot topic of debate.

In practice, central bank independence is categorized differently by various experts. Fraser (1994), DeBelle and Fischer (1995), and Meyer (2000) divided central bank independence into goal independence and instrument independence. Goal independence means the central bank itself sets the goals to be achieved, while instrument independence implies that the central bank has sufficient jurisdiction to implement its duties in order to achieve the goals set. On the other hand, Grilli, Masciandaro, and Tabellini (1991) and Elgie (1995) divided central bank independence into political independence and economic independence. Political independence implies that the central bank has the freedom to set its goals or policy decisions free from government interference, while economic independence means that the central bank is free to use all policy instruments at its disposal, without government restrictions, to achieve the goals.¹

Baka (1994) and Mboweni (2000) specifically added another aspect of independence, namely financial independence, implying that the central bank has access to adequate financial resources and exerts full control over its budget. In general, financial independence is not always considered when assessing the variables of independence despite several studies evidencing a positive relationship between financial independence and central bank performance and policy credibility. Other categories have been developed, which generally contain nearly the same aspects, together with the advantages and disadvantages of each.

Bofinger (2001) classified three categories as a synthesis of the above, namely: (1) goal independence, requiring that the government has no direct influence when setting the goals of monetary policy; (2) instrument independence, requiring that

¹Political independence also extends to nomination, recruitment, dismissal, qualifications, tenure, reappointment, and other officers than the governor, deputy governor, and members of the BOGs as well as the decision-making process, which encompasses decisionmaking, directives from the government, veto from government representatives, BOG remuneration, and central bank capital ownership.

the central bank can autonomously set the operational targets of monetary policy; and (3) personal independence, requiring that the decision-making body of the central bank exists without formal direction or informal pressure from the government.²

11.2.1.1. Goal Independence. Fundamentally, the goal of monetary policy can be defined in broad terms, not only as the choice between price stability and output, but also the target horizon, the concrete indicators used, the value of the target and the escape clauses.³ Therefore, goal dependence is reflected by different conditions. In this case, a central bank could enjoy independence over the full scope of monetary policy goals, while another central bank may enjoy independence over several aspects only.

The central bank of the United States, the Federal Reserve (the Fed), owns the highest degree of goal independence among central banks around the world. The mandate of the Fed is formulated in broad terms, therefore the work activities relate to economic growth and creating job opportunities. By mandating the achievement of several monetary policy goals for the Federal Reserve System and Federal Open Market Committee (FOMC) through prevailing laws, without allocating a specific weight for each policy goal, the Fed owns full independence to determine its preferences, for instance, whether to seek price stability or create more job opportunities.

In contrast, the European Central Bank (ECB) is considered to have a more limited degree of goal independence, which is evidenced by the emphasis on price stability as the monetary policy goal mandated by law. Nevertheless, considering the formulation of goal achievement does not require a horizon, the ECB is free to choose whether to achieve the target in the near term or over a longer horizon. In terms of political aspects, prevailing regulations explicitly proscribe political interference from governments, parliaments, or the EU Commission in ECB task implementation. Such goal independent regulations are also found in other countries, including Japan and Sweden.

Central banks with low goal independence include the UK, Canada, and New Zealand. In those countries, the overriding goal of monetary policy, as mandated by prevailing laws, is the achievement of price stability but the discrete value of the target is in the hands of the respective government. In New Zealand, price stability is defined in the Policy Target Agreements (PTA) negotiated periodically between the Minister of Finance and Central Bank Governor. In the UK, the law stipulates that the Minister of Finance is responsible for defining and publishing what is meant by price stability. In Canada, however, the central bank and federal government jointly announce the inflation-control target.

Going beyond the degree of independence experienced by the central banks listed above, the literature also supports central banks that are not goal

²Quantitatively identifying the degree of independence, in the form of an independence index, has been achieved in several studies, including Cukierman (1992) and Bofinger (2001).

³A clause in a contract that specifies the conditions under which one party can be freed from an obligation.

independent (Bernanke et al., 1999; Fischer, 1995) because the final policy goal must fundamentally reflect the will of the people and, thus, should be set by officials elected representatively. Notwithstanding, Lohmann (1992) stressed that placing trust in elected officials could also prompt the risk of inflation bias because the government, or parliament, has a short-term orientation, hence its influence on monetary policy implementation could trigger shocks that exacerbate inflationary pressures. In addition, short-term supply-side shocks also encourage the government, or parliament, to tolerate deviation between target achievement and the medium-term target set.

Consequently, there is a trade-off between monetary policy flexibility in response to supply-side shocks (with full commitment to achieving price stability) and the degree of independence (politically). Experience from the central bank of Germany, the Bundesbank, which is known as a particularly conservative central bank, showed that the risk of inflation bias was considered more relevant than the risk of an independent central bank that is unwilling to accommodate short-term supply-side shocks. The Bundesbank, for example, prefers to respond to supply-side shocks more flexibly. In fact, the Bundesbank was open to an implicit inflation target, set at 2%, for a number of periods.

11.2.1.2. Instrument Independence. Instrument independence implies that the central bank sets its own operational targets without government intervention. According to Bofinger (2001), independence has three important control elements as follows: (1) controlling short-term interest rates as the main operational target; (2) controlling the exchange rate as a supplementary operational target in the context of an open economy; and (3) credit restrictions from the central bank to the government, which could undermine control of the monetary base and, similarly, short-term interest rates. It is important to note that the first element assumes the propensity of the central bank is generally toward using short-term interest rates, rather than the monetary base (base money), as the main operational target.

In practice, monetary policy implementation by the central bank can be directed autonomously to control interest rates. Nonetheless, as in the case of goal independence in several countries, such as the UK, Canada, and New Zealand, the government can intervene in the decisions taken by the central bank. With respect to controlling exchange rates, there are currently no central banks in the world with unrestricted authority to regulate exchange rates. The ECB owns broad jurisdiction, where prevailing laws have mandated authority to formally regulate the exchange rate system for members of the European Council (consisting of the Ministers of Finance of the member nations of the European Union), after consultation with the European Parliament. A more limited mandate would be given if a formal exchange rate system was not strived for, namely if the European Council could formulate general exchange rate policy orientation after receiving ECB recommendations. Differing from conditions at the ECB, other central banks generally receive limited exchange rate policy authority, which is typically mandated to the government.

Concerning lending restrictions to the government, central bank laws generally regulate specific clauses. Despite the possibility of disbursing loans to the

government, depending on the economic conditions and political dynamics in each respective country, most central banks are restricted in terms of lending to the government because the impact could lead to accelerating inflation. In general, however, the central bank is guaranteed to achieve the monetary policy goals if politically independent and given the authority to decide when and how much to lend to the public sector.

11.2.1.3. Personal Independence. Despite enjoying goal independence and/or instrument independence, a central bank may struggle to achieve personal independence because the government could exert informal pressures on monetary policy implementation. For example, the governor of the central bank could be removed at any time without a firm reason or the top of the central bank's leadership structure may only enjoy the support of one or two individuals, thus the government could appoint an individual with the same common interests as the government. Therefore, observing the degree of personal independence at a central bank involves two main elements, namely the number of members serving on the highest decision-making forum (Board of Governors (BOG) for instance) and length of tenure of each respective member. More BOG members, or longer tenures, imply a greater degree of independence from government intervention.

The following table presents the conditions of personal independence at several central banks based on the two elements mentioned. In general, compared to other central banks, the US Federal Reserve, and ECB benefit from the greatest degree of independence. In contrast, the Reserve Bank of New Zealand (RBNZ) has the lowest degree of independence.

Empirically, there are several different methods to measure the level of independence at a central bank in comparison to other central banks and over time. Experts have identified and developed independence indicators used to subsequently measure the degree of independence at a central bank. For example, Parkin (1978, 1987) developed an approach to differentiate the various types of central bank independence. Burdekin, Clas, and Thomas (1992) as well as Masciandaro and Spinelli (1994) developed an approach to rank independence. Meanwhile, Cukierman, Webb, and Neyapti (1992) developed a precise measure of central bank independence, using 16 indicators grouped into eight indicator sets, which were weighted between 0.20 for the four indicator sets concerning the government and 0.05 for the indicator sets containing one indicator regarding loans disbursed by the central bank.

The method to measure the degree of personal independence employed by Robert Elgie (1995), who refined the method of Cukierman et al., shall be explored in more depth. Elgie used 29 indicators of political independence, compared to just five used by Cukierman et al. Furthermore, Elgie also used seven indicators for the governor, compared to just four per Cukierman et al. Finally, Elgie used five indicators for the internal decision-making process, eight indicators for the deputy governor and nine for the BOGs.

Concerning economic independence, Elgie only used seven indicators, while Cukierman et al. used 11. Eight of the 11 were linked to central bank restrictions on lending to the government, however, which Elgie combined into just one indicator. In general, Elgie developed a group of economic independence indicators

Table 11.1: Central Bank Personal Independence.

Central Bank	Highest Decision-making Forum	Total Members	Tenure
Europe	Governing Council	6 Executive Board Members, 11 Central Bank Governors from member states	Min 5 years: Central Bank Governors from member states Min 8 years: Executive Board Members
United States	Federal Open Market Committee	7 BOG members, 5 Presidents of Federal Reserve Banks	7 years: BOG members
United Kingdom	Monetary Policy Committee	1 Governor, 2 Deputy Governors, 6 special members	5 years: Governor and Deputy Governors 3 years: Directors
Canada	Executive Committee (Governing Council)	1 Governor, 1 Deputy Governor, 5 Directors	5 years: Governor and Deputy Governor 3 years: Directors
Japan	Policy Board	1 Governor, 2 Deputy Governors, 6 special members	5 years
New Zealand	Governor	Governor	5 years

Source: Bofinger (2001).

that provided a more comprehensive picture. Moreover, the refined methodology consists of far more comprehensive political and economic independence indicators.

11.2.2. Central Bank Accountability

Congruent with the degree of independence enjoyed, in terms of increasing credibility, a central bank must be held publicly accountable and clarify monetary policy implementation. In general, accountability can be observed from two aspects. *First*, from the single goal that makes the central bank more trustworthy because it clarifies the measures and accountability. *Second*, the appointment process and short tenure of the central bank governor as well as possible reappointment could improve accountability because it would allow for performance appraisals.

In practice, central bank accountability could also be observed from the definition of accountability per de Haan, Amtenbrink, and Eijffinger (1999). The following table presents a comparison of central bank accountability in

several countries based on at least 13 aspects linked to three characteristics of accountability, namely the final policy goal, ultimate accountability, and political transparency.

From [Table 11.2](#), we can see that, if each aspect with a “Yes” condition is given a weight of 1 (one), it could be concluded that the Bank of England has the highest degree of accountability. Legally, the ECB is seen to have the lowest degree of accountability. It is important to note that, in practice, for several aspects (3, 4, and 11), the ECB has applied principles far beyond those mandated by central banking laws. Consequently, in practice, the degree of ECB accountability is higher than just based on prevailing laws. Per Issing (1999), however, such conditions do not make the ECB “among the most transparent and accountable central banks in the world.”

11.2.3. Correlation between Independence and Accountability

Nowadays, people believe that central bank independence could encourage price stability. Therefore, achieving central bank independence is not the final goal but a means to achieve the policy objectives stipulated in law. Providing a mandate that requires accountability implies that the central bank must apply sound accountability principles to the policy-making process, meaning the central bank must inform and justify several major aspects, such as the progress of achieving the policy goals, the monetary policy strategy and current policy measures. How far, therefore, can an independent central bank meet those requirements?

Answering that question, de Haan, Amtenbrink, and Eijffinger (1999) conducted an empirical study linking central bank independence and accountability in advanced countries. An accountability index was created by scoring each of the 13 aspects covering the three characteristics of accountability, namely the final goal, ultimate accountability, and transparency. Meanwhile, an accountability index was also developed based on the assessments performed by Eijffinger and Schaling (1993) as well as Eijffinger and Van Kuelen (1995), per three aspects, namely the weight of holding the ultimate accountability for monetary policy (central bank/government/central bank and government), the presence of government representatives on the BOGs (yes/no), as well as the fraction of board members appointed by the government (more/less than half). The total independence and accountability index is presented in [Table 11.3](#).

Based on the data, the linkages were assessed using a regression approach between the independence index and each respective accountability characteristic index and total accountability index. The assessment produced several observations as follows:

- (1) There is a negative correlation between independence and two characteristics of accountability, namely ultimate accountability and transparency.
- (2) There is a positive but insignificant correlation between independence and ultimate accountability.
- (3) In general, there is a negative correlation between independence and accountability, albeit not as strong as the respective correlation in point (1).

Table 11.2: Central Bank Accountability.

Accountability Aspects	Europe	United States	United Kingdom	Canada	Japan
<i>(A) Final Policy Goal</i>					
(1) Stipulated in the central banking act	Yes	Yes	Yes	Yes	Yes
(2) Goal prioritization	Yes	–	Yes	–	–
(3) Clear goal definition	– (Yes)	–	Yes	–	–
(4) Goal quantification (per prevailing laws)	– (Yes)	–	Yes	–	–
<i>(B) Ultimate Accountability</i>					
(5) Monitored by parliament	Yes	Yes	Yes	Yes	Yes
(6) Government right to provide instructions	–	–	Yes	Yes	Yes
(7) Procedures to implement a policy rejection mechanism	–	Yes	Yes	Yes	Yes
(8) Possibility to request government instruction	–	–	–	–	–
(9) Possibility of amending laws by parliament majority	–	Yes	Yes	Yes	Yes
(10) Using past performance to consider removal of central bank governor from office	–	–	–	–	–
<i>(C) Political Transparency</i>					
(11) Required to publish a monetary policy report (inflation report)	– (Yes)	Yes	Yes	Yes	–
(12) Publishes minutes of the Board meeting	–	Yes	Yes	–	–
(13) Required to provide public clarification if the target is achieved	Yes	Yes	Yes	Yes	Yes

Source: de Haan et al. (1999).

Table 11.3: Central Bank Independence and Accountability Index.

Country	Independence	Accountability		
		Final Goal	Ultimate Accountability	Transparency
Australia	1	1	5	1
Belgium	3	0	3	0
Canada	1	1	4	2
Denmark	4	1	2	1
France	4	3	2	1
Germany	5	2	1	0
Italy	2	0	3	1
Japan	3	1	4	1
Netherlands	4	1	4	0
New Zealand	3	4	4	2
Spain	3	3	2	2
Sweden	2	0	3	2
Switzerland	5	1	1	0
UK	3	4	4	3
US	3	1	2	3
ECB	5	3	0	1

Source: de Haan et al. (1999).

In general, the results showed an inverse correlation between central bank independence and accountability. Nevertheless, the results do not imply that it would be impossible to simultaneously achieve independence and accountability. The general belief at central banks that independence and accountability can be strived for together is also very strong, in line with the prevailing opinion that accountability, in terms of democratic accountability, should not be seen as a mechanism that limits independence at a central bank; but should be seen as a mechanism to reinforce the legitimacy of central bank independence. In other words, even if it is difficult to consider as a requirement, accountability should be considered complementary to independence.

11.3. Theoretical Models Concerning Independence and Accountability

Several studies have explored the theoretical evidence and conducted empirical testing regarding the importance of institutional development in terms of central bank policy implementation. Nonetheless, in contrast to the earlier and already established studies that focused on aspects of independence, studies on aspects of

accountability remain at a relatively embryonic stage.⁴ The following section presents the evolution of theory concerning independence, or delegation of authority, and central bank accountability.

11.3.1. *Delegation of Authority Theory: Conservative Agent*

In one of his seminal pieces, Kenneth Rogoff (1985) stated that an independent and inflation-averse central bank would tend to lower average inflation but increase output variability. In other words, a conservative central bank could reduce inflation bias caused by the time inconsistency problem but also play a more limited role in terms of output stability.

This was explained through a model popularized by Barro and Gordon (1983), which can generally be developed as follows. Output performance (y) is formulated based on the supply curve specified à la Lucas:

$$y_t = a(\pi_t - \pi_t^e) + \varepsilon_t \quad (1)$$

where π_t is inflation, π_t^e is inflation expectations, and ε_t is independent and identically distributed (iid) shocks with an average of 0 and variance σ_ε^2 . a reflects the influence of fallacies on output. Expectations are formed before the presence of shocks and before policymaker determine π_t .

The policymakers' objective function is:

$$L_t = \pi_t^2 + b(y_t - y^*)^2 \quad (2)$$

where, $b > 0$ and the natural level of output, $y^* > 0$. Substituting (1) for (2) produces:

$$L_t = \pi_t^2 + b(a(\pi_t - \pi_t^e) + \varepsilon_t - y^*)^2 \quad (2')$$

The first-order condition minimizes the loss function as follows:

$$\pi_t = \left(\frac{aby^*}{1+a^2b} \right) + \left(\frac{a^2b}{1+a^2b} \right) \pi_t^e - \left(\frac{ab}{1+a^2b} \right) \varepsilon_t \quad (3)$$

⁴After the seminal theoretical contribution of the “conservative central bank” (Rogoff, 1985), the literature on central bank independence was enriched by other studies, including “insulation from political uncertainty” (Alesina & Gatti, 1995) and “optimal contract” (Walsh, 1995). Meanwhile, a prominent theoretical argument on accountability is contained in “democratic accountability” (Eijffinger, Hoeberichts, & Schaling, 1998).

Assuming $\pi_t^e = E_{t-1}(\pi_t)$, produces:

$$E_{t-1}(\pi_t) = aby^* > 0 \quad (4)$$

$$\pi_t = aby^* - \left(\frac{ab}{1+a^2b} \right) \varepsilon_t \quad (5)$$

$$y_t = \left(\frac{1}{1+a^2b} \right) \varepsilon_t \quad (6)$$

Policy rule (5) contains the component known as inflation bias aby^* and component of stabilization $(ab/(1+a^2b))\varepsilon$. From equations (4)–(6), the following unconditional moments are obtained:

$$E(\pi_t) = aby^* \Rightarrow \frac{\partial E(\pi_t)}{\partial b} > 0 \quad (7)$$

$$E(y_t) = 0 \quad (8)$$

$$\text{var}(\pi_t) = \left(\frac{ab}{1+a^2b} \right)^2 \Rightarrow \frac{\partial \text{var}(\pi_t)}{\partial b} > 0 \quad (9)$$

$$\text{var}(y_t) = \left(\frac{1}{1+a^2b} \right)^2 \Rightarrow \frac{\partial \text{var}(y_t)}{\partial b} < 0 \quad (10)$$

From the results above, it could be concluded that monetary policy based on discretion would cause high inflation without stimulating output beyond its natural level, also known as stagflation. In addition, there is a stabilization trade-off, where a smaller weight of output stabilization (b) would lead to lower and more stable inflation, but accompanied by output volatility. In this case, inflation bias could be overcome through future (*ex ante*) commitment from policymaker to abide by an optimal policy rule, depending on the realization of ε . Realistically, such commitment is not a choice available to policymaker due to difficulties identifying the characteristics of the shocks, thus efforts to eliminate market distortions and frictions could incur a large cost.

Therefore, to approach optimal social results, as if there were commitment from the policymaker, the practical solution is the delegation of authority. In this case, Rogoff suggested that public prosperity could be improved if the policymaker delegate the (*ex ante*) choice of monetary policy to a previously selected independent agent. Independent, in this case, means that the agent cannot be removed *ex post*, when determining the policy choice. The policymaker could, however, select an agent with different preferences (public preferences), reflected in parameter b . System behavior is each respective period is as follows: First, the policymaker will select an agent from the population with parameter b_t , then, after the expectations have been formed and the shocks have occurred, the agent determines policy.

Meanwhile, the agent's preferences are formulated as follows:

$$L_t = \pi_t^2 + \hat{b}_t (y_t - y^*)^2 \quad (11)$$

If a certain agent is selected with parameter b_j , then inflation and output shall form as follows:

$$\hat{\pi}_t = a\hat{b}y^* - \left(\frac{a\hat{b}}{1+a^2\hat{b}} \right) \varepsilon_t \quad (12)$$

$$\hat{y}_t = \left(\frac{1}{1+a^2\hat{b}} \right) \varepsilon_t \quad (13)$$

In each subsequent period, the selected agent may be different but because the behavior in each period is identical, the optimal choice of the policymaker (and the public, if aligned) is as follows:

$$\text{Min}_b E(L(b, \hat{b})) = E(\hat{\pi}_t^2 + b(\hat{y}_t - y^*)^2) \quad (14)$$

$$= E \left\{ \left[a\hat{b}y^* - \left(\frac{a\hat{b}}{1+a^2\hat{b}} \right) \varepsilon_t \right]^2 + b \left[\left(\frac{1}{1+a^2\hat{b}} \right) \varepsilon_t - y^* \right]^2 \right\} \quad (15)$$

$$= (a\hat{b}y^*)^2 + \left(\frac{a\hat{b}}{1+a^2\hat{b}} \right)^2 \sigma_\varepsilon^2 + b \left(\frac{1}{1+a^2\hat{b}} \right)^2 \sigma_\varepsilon^2 + by^{*2} \quad (16)$$

Equation (15) shows that when selected, the agent will follow rule (5) with preference parameter b . Considering the policymaker acts earlier, before π_t^e and ε_t have been realized, the policymaker sets expectations based on the shocks that have occurred.

The first-order condition of the solution to the minimization of the loss function problem is as follows:

$$\frac{\partial L}{\partial \hat{b}} = 0 \Rightarrow \left(\frac{y^*}{\sigma_\varepsilon^2} \right) (1+a^2\hat{b})^3 \hat{b} = b - \hat{b} \quad (17)$$

Such conditions show the presence of parameter b in the search for solution $f(b) = b - \hat{b}$, which can be represented graphically as shown in Fig. 11.1 (Walsh, 2001):

The solution is therefore:

$$0 < \hat{b} < b \quad (18)$$

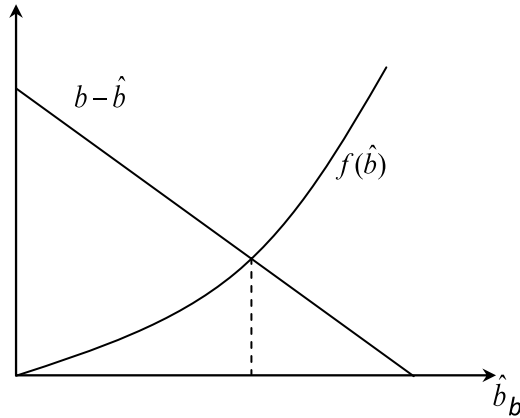


Fig. 11.1: Output Stabilization Preference Behavior.

From the solution, we can conclude that the policymaker could improve welfare (and therefore public welfare if aligned) by delegating monetary policy to a more inflation-averse or conservative agent than the policymaker itself. By analogizing that the policymaker is represented by the government or parliament, and the agent is represented by the decisionmaker and accountability of the central bank, the fundamental implications of the delegation of authority theory are as follows. In comparison to countries where the central bank is not independent, independent central banks will experience:

- (1) lower and more stable average inflation; and
- (2) higher output volatility as a consequence of efforts to control inflation, but average output will not change.

11.3.2. Central Bank Accountability Theory: Democratic Accountability

In reality, the delegation of authority, in terms of monetary control, by the government to the central bank is not without risk. Therefore, there should be a mechanism to guarantee that monetary policy implementation by the central bank is aligned with the desires/hopes of the public. In other words, the central bank must demonstrate sound accountability to the institutions democratically elected by the public. How to understand the meaning of accountability in a democratic society?

In a study on the issue of accountability, de Haan et al. (1999) differentiated three main characteristics in the meaning of central bank accountability, namely: (1) Who takes the decision on the final goal of monetary policy? (2) Who is ultimately accountable for monetary policy implementation? (3) How politically

transparent is the monetary policy?⁵ Eijffinger et al. (1998) developed a theoretical model concerning central bank accountability, focusing their analysis on two characteristics of accountability, namely political transparency and ultimate accountability.⁶ Meanwhile, the issue of determining the final goal is considered in relation to the goal independence of the central bank.

The theoretical model of accountability was developed based on the studies by Lohmann (1992) as well as Nolan and Schaling (1996). The model scenarios are as follows:

- (1) The government has delegated monetary policy implementation to a conservative central bank. Nevertheless, the government and public are not exactly sure the central bank's relative preferences to price stabilization and output.
- (2) Subsequently, after the central bank has proposed its inflation preferences, the government could decide to review or reject the proposal, thus incurring a certain cost and, therefore, the government could set the inflation target. In this case, the central bank is considered partially independent.

The cost of rejecting the proposal relates to the question of who is most accountable for monetary policy implementation. If the government is unwilling to bear the cost, the central bank is considered the most accountable for monetary policy implementation. In contrast, if the cost is considered small, the government is held accountable for monetary policy implementation.

In concrete terms, the theoretical model developed can be expressed in the following way, with output determined by the supply function à la Lucas and simplified as follows:

$$y = \pi - \pi^e + \varepsilon, \text{ dengan } \varepsilon \sim N(0, \sigma_\varepsilon^2) \quad (19)$$

⁵Geraats (2000) differentiated five aspects of transparency that support accountability, namely political, procedural, policy, operational, and economic transparency. Political transparency encompasses formal goals, quantitative targets, and clarification of institutional structure. Furthermore, political transparency is the most important aspect because it provides the criteria and identifies who is accountable. Economic, procedural, and policy transparency are required to understand the background/motivation behind the policies instituted, while operational transparency is required to comprehend the constraints to successful policy achievement.

⁶Accountability and transparency are inextricably linked. The central bank must be held accountable for task implementation to its stakeholders. A more transparent institution shall also increase central bank accountability and, ultimately, boost central bank performance through greater market efficiency and increased decision-making clarity (Poole, 1999, 2003). Meanwhile, better performance would improve accountability at the institution/central bank involved. Nevertheless, transparency is a necessary condition for accountability but not a sufficient condition because accountability is also determined in terms of monetary policy implementation.

The objective function of the government is:

$$L_G = \frac{1}{2}\pi^2 + \frac{1}{2}(y - y^*)^2 + \delta c \quad (20)$$

where $y^* > 0$ is the government's target output. If the government rejects the inflation target proposed by the central bank, the government will bear the cost, c , determined by the prevailing political institutions. δ is a dummy variable with a value of 1 if the government rejects the central bank's proposal and 0 if the government accepts the central bank's proposal.

Meanwhile, the central bank has random (stochastic) preferences, where the degree of conservatism is calculated in the contract (quadratic function) with parameter f . In addition, the central bank has private information concerning the realization of shocks in its preferences, with a uniform distribution. Therefore, the loss function of the central bank is formulated as follows:

$$L_{CB} = \frac{1-x}{2}\pi^2 + \frac{1}{2}(y - y^*)^2 + \frac{f}{2}\pi^2, \quad (21)$$

with $x \sim U[-h, h]$ and $h < f$.

Without delegating monetary policy, the government sets (with discretion) an inflation target that minimizes the loss function, namely:

$$\pi_G = \frac{y^* + \pi^e - \varepsilon}{2} \quad (22)$$

If monetary policy has been delegated to the central bank, the inflation target is set based on minimizing the loss function of the central bank, namely:

$$\pi_{CB} = \frac{y^* + \pi^e - \varepsilon}{2 - x + f} \quad (23)$$

Considering that the central bank is more conservative, or inflation-averse, than the government ($f - x > 0$), the policy preferences of the central bank would feed through to lower inflation bias than from the government but, on the other hand, would also be less active in response to supply-side shocks.

If the government decided to reject the inflation target proposed by the central bank (23), however, the government would then use its own inflation target (22). Therefore, the loss borne by the government would be ($\delta = 1$):

$$L_G(\pi_G) = \frac{1}{4}(y^* + \pi^e - \varepsilon)^2 + c \quad (24)$$

Meanwhile, if the government decides to accept the central bank's proposed inflation target, the loss function would therefore be ($\delta = 0$):

$$L_G(\pi_{CB}) = \frac{1 + (x - 1 - f)^2}{2(x - 2 - f)^2}(y^* + \pi^e - \varepsilon)^2 \quad (25)$$

The two choices show whether the government rejects or accepts the central bank's proposal, namely if:

$$L_G(\pi_G) < L_G(\pi_{CB}) \quad (26)$$

If the government decides the cost of rejecting the central bank's proposal outweighs the benefits of the inflation target set by the government, the central bank will be independent. By substituting equations (24)–(26), we can conclude that the central bank will be independent if:

$$c > \frac{(x-f)^2}{4(x-2-f)^2} (y^* + \pi^e - \varepsilon)^2 \quad (27)$$

Therefore, depending on shock realization, the central bank would face two conditions upon acting, namely independence $((x, e) \in I)$ or accommodative $((x, e) \in A)$. If the central bank is accommodative, the inflation target would represent the weighted average of inflation based on the government's preference and inflation based on the central bank's preference, as follows:

$$\pi_{ACC} = \varphi\pi_G + (1-\varphi)\pi_{CB} = \frac{2-\varphi(x-f)}{2(2-x+f)} (y^* + \pi^e - \varepsilon)^2 \quad (28)$$

Accordingly, substituting [equation \(28\)](#) for [\(20\)](#) and using [equation \(19\)](#) shows that the loss incurred by the government is as follows:

$$L(\pi_{ACC}) = \frac{1+(x-1-f)^2 - \varphi(x-f)^2(1-\frac{1}{2}\varphi)}{2(2-x+f)^2} (y^* + \pi^e - \varepsilon)^2 \quad (29)$$

The central bank will always take accommodative actions; hence the government is indifferent to rejecting the central bank's proposal. Therefore, the central bank will have ϕ , thus $L_G(\pi_{ACC}) < L_G(\pi_G)$. By combining [equations \(20\)](#) and [\(29\)](#):

$$\varphi = 1 - \frac{2(2-x+f)\sqrt{c}}{|y^* + \pi^e - \varepsilon| |x-f|}, \text{ if } (x, \varepsilon) \in A, \text{ and } \phi = 0 \text{ if } (x, \varepsilon) \in I \quad (30)$$

By substituting [equation \(20\)](#) for [\(28\)](#), we obtain a rate of inflation if the central bank implements accommodative measures, namely:

$$\pi_{ACC} = \frac{y^* + \pi^e - \varepsilon}{2} - \sqrt{c} \operatorname{sgn}(f-x) \operatorname{sgn}(y^* + \pi^e - \varepsilon) \quad (31)$$

Assuming that the central bank will always act conservatively irrespective of the shocks occurring ($f > x$), then [equation \(31\)](#) could be expressed as follows:

$$\pi_{ACC} = \frac{y^* + \pi^e - \varepsilon}{2} - \sqrt{c} \operatorname{sgn}(y^* + \pi^e - \varepsilon) \quad (32)$$

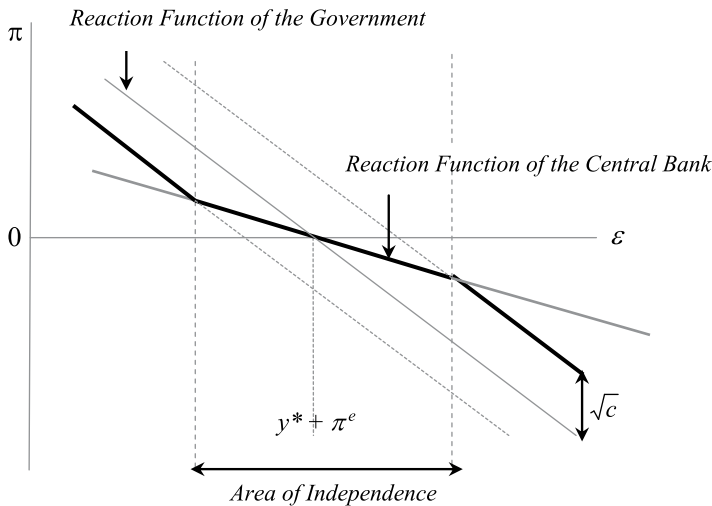


Fig. 11.2: Monetary Policy Behavior ($c > 0$).

Based on that interpretation, the actual behavior of monetary policy, with the rejection cost, can be explained by Fig. 11.2.

We can explain that the area around $y^* + \pi^e = \varepsilon$ is the area where the central bank is independent in terms of setting the inflation target per its own preferences. Outside of that area is where the central bank has to accommodate an inflation target based on the preferences of the government.

In this area, the government sees that the sacrifice ratio of stabilization measures due to supply-side shocks is large, therefore an inflation target set by the central bank would be considered insufficient. Parallel to the government's reaction function are two reaction functions that occur due to the cost of rejecting the central bank's inflation proposal. The intersection between that reaction function and the reaction function of the central bank determines the area of independence.

Based on such behavior, we can observe the implications of one of the characteristics of accountability, namely responsibility, on economic developments, particularly the impact on inflation and shock stabilization.⁷ In this case, one way to improve central bank accountability is to delegate (technically, a shift in direction) the burden of responsibility for monetary policy from the central bank to the government. In the theoretical model, this can be achieved by reducing the rejection cost (c).

Fig. 11.3 shows that the distance between the government's reaction function and two parallel lines is narrowing and, consequently, the area of central bank independence is also narrower. We can see that a lower rejection cost: (1)

⁷A more in-depth discussion on the aspects of transparency is presented in the following chapter on Policy Transparency and Communication Strategy.

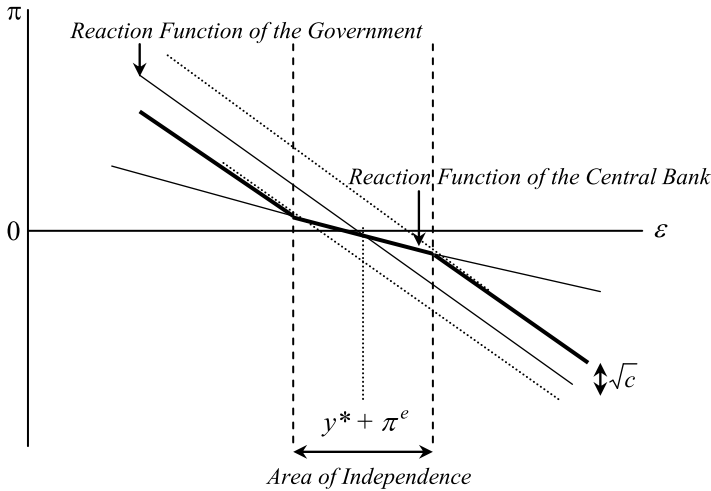


Fig. 11.3: Monetary Policy Behavior after Lowering c .

undermines central bank independence; (2) raises inflation expectations; and (3) increases stabilization to shocks. Therefore, increasing accountability by delegating the burden of responsibility of monetary policy from the central bank to the government is especially applicable to countries suffering from a relatively serious flexibility problem (high σ_E^2) compared to a credibility problem (high y^*).

Meanwhile, another aspect of accountability, namely transparency, can be explained by its correlation with the element of uncertainty in the behavior (preferences) of the central bank. In this case, increasing the degree of transparency at a central bank relates closely to reducing the degree of uncertainty in the central bank's preferences (lowering the value of h in equation (21)). Lowering the degree of uncertainty would make the central bank more conservative, as reflected in the reaction function of the central bank, which becomes flatter, therefore, central bank transparency would also lower inflation expectations. Another consequence of lowering the degree of uncertainty (lowering the value of h) would be to reduce the accommodative nature of the central bank to efforts to stabilize shocks (Fig. 11.4).⁸

Therefore, increasing accountability by delegating the burden of responsibility for monetary policy from the central bank to the government is especially applicable to countries suffering from a relatively serious credibility problem (high y^*) compared to a flexibility problem (high σ_E^2). It is important to note that although the area of independence narrows, increasing accountability through greater transparency does not effectively undermine central bank independence because greater transparency ensures the macroeconomic benefits obtained by the central bank are aligned with the central bank's preferences.

⁸Mathematically, the reduction can be seen in Eijffinger and Hoeberichts (2000).

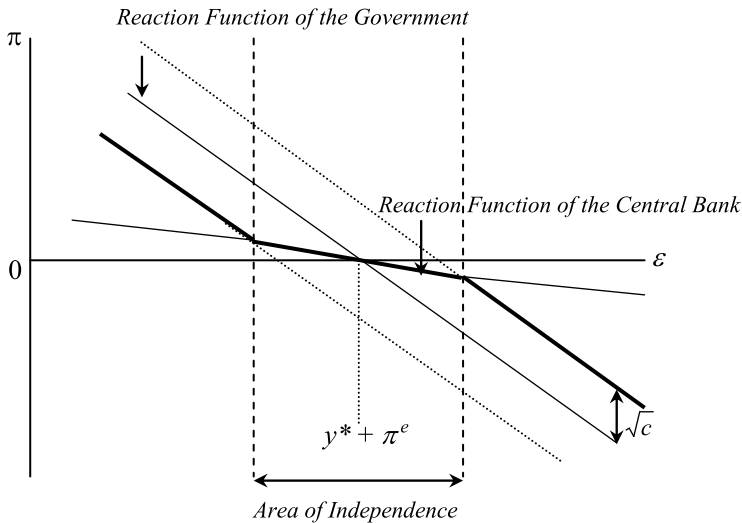


Fig. 11.4: Monetary Policy Behavior after Lowering h .

11.4. Empirical Studies and Related Issues

11.4.1. Central Bank Independence and Economic Performance

There are numerous studies on central bank independence that support or reject the benefits of central bank independence based on different arguments. Those who support the benefits of independence are based on the evidence that:

- (1) Greater central bank independence leads to lower and more stable inflation, which can drive long-term economic growth.
- (2) Greater central bank independence leads to a lower long-term budget deficit due to a separation of authority between the printing and spending of money.

Those who reject the benefits of independence are based on the evidence that:

- (1) Although there is a correlation between independence and low inflation, it does not mean that through greater independence, lower inflation will be achieved.
- (2) Monetary policy is a part of overall economic policy, therefore, there is no reason to separate fiscal policy, monetary policy, trade policy, and other policies.
- (3) If the central bank officers are not appointed democratically, then the decisions taken on interest rates, exchange rates, inflation and other monetary issues are not representing the interests of the public. Consequently, independence could create an abuse of authority.

Regardless of the differing opinions, the reality is that more and more central banks in advanced countries and developing countries are becoming independent.

Several studies have been conducted to observe the empirical linkages between independence and economic performance achieved by the central bank, including Grilli et al. (1991), Cukierman (1992), Alesina and Summers (1993), and Bofinger (2001). Grilli et al. (1991) concluded that political and economic independence have an inverse correlation to inflation. Meanwhile, several other forms of independence correlate but to a varying degree, implying that by giving the central bank independence, lower inflation could be achieved. Grilli et al. also concluded that political and economic independence do not have a significant correlation with real output growth. In general, therefore, central bank independence could result in stronger economic performance, in the form of lower inflation, without sacrificing economic growth.

While that conclusion has been corroborated by a number of other studies, the research conducted by Alesina and Summers (1993) produced a rather different conclusion. Alesina and Summers concluded that there was a significant inverse correlation between central bank independence and average inflation in western industrialized nations from 1955 to 1988. Nevertheless, no symmetrical correlation was found between central bank independence and average economic growth.

It is important to note that the study used relatively old data (excluding the 1990s). Bofinger (2001) performed a similar study using an independence index based on three elements, namely goal, instrument, and personal independence.⁹ That study produced a conflicting conclusion to the previous research, namely that central bank independence did not correlate with the achievement of low inflation in advanced countries during the period from 1996 to 2000. The surprising results were linked to the comparatively brief observation period, thus demanding some caution. *First*, central banking laws that emphasis aspects of independence were first promulgated in the middle of the 1990s, therefore, observations and comparisons between central banks over a longer horizon are not possible. *Second*, the period from 1996 to 2000 was a time when central bank policy around the world was oriented toward achieving price stability and, thus, is often considered the base period to start comparisons of central bank performance.

The conceptual problem of conflicting conclusions in the existing empirical research on the linkages between independence and inflation fundamentally lies in the condition that central bank independence is ideally required in situations where the government strives to prioritize inflationary policy in order to reduce unemployment. Therefore, a politically independent central bank will only create inflation bias if:

⁹It is important to note that the independence index can be estimated based on several criteria (elements of independence) by calculating the weight of each respective criterion. Subjectively, therefore, using an independence index can be different between different studies.

- (1) The economy is facing a serious unemployment problem.
- (2) The inflation problem is not taken seriously enough.
- (3) There is a short-term political agenda to maintain power in a subsequent term.

In other words, countries that maintain a low level of unemployment, prefer price stability, and enjoy political stability should have a negligible inflation bias, even with a central bank that is not independent.

Such conditions were personified by the economic cases of Japan and France in the 1980s and 1990s, when policy was oriented toward price stability, while neither central bank was independent until 1993 (Banque de France) and 1998 (Bank of Japan). With the current policy preference oriented toward price stability, the implication is that central bank independence has become less critical. Nevertheless, the empirical evidence showed a potential inflation spike during periods of low inflation, for instance during the 1950s and 1960s. Therefore, mandating central bank independence is considered a strategic preventative measure for sound monetary policy implementation.

11.4.2. Related Issues

11.4.2.1. The Critical Issue of Central Bank Independence: Two Fallacies. There are at least two conclusions fundamental to the spectrum of studies on central bank independence, including the works of Rogoff (1985), Persson and Tabellini (1993), Alesina and Gatti (1995), and Walsh (1995). *First*, the lack of (pre) commitment in policy implementation by an independent central bank would not avoid discretionary policy that leads to inflation bias. *Second*, a contract submitted by the government to the central bank could overcome the time inconsistency problem, which is the root of inflation bias.

The conclusions above received criticism from McCallum (1995), who showed two fallacies in the interpretations, the first being inconsistencies between the analysis and reality. In this case, the range of studies on central bank independence tended to underestimate the sound monetary policy achievements of an independent central bank. In addition, there were general errors when interpreting the beneficial effects of the contracts made by the central bank.

Several studies have concluded, however, that when determining the optimal policy response, the central bank is faced with two alternative choices in the form of a solution based on discretion and a solution based on a rule.¹⁰ According to McCallum, there is no justification for the opinion which assumes that, with commitment, an independent central bank will tend to act based on discretion rather

¹⁰Compared to setting inflation on a period-by-period basis or based on discretion, the central bank bases inflation on a pre-existing policy rule when the inflation expectations are formed.

than a policy rule.¹¹ In practice, there are no meaningful impediments to a central bank acting based on a policy rule, which facilitates better results. Furthermore, even without pre-commitment as the basis of assured future measures, this does not mean that the policy measures instituted by the central bank would not be feasible for implementation.

Therefore, McCallum continued to recommend the adoption of a policy rule by central banks. It is important for central banks to avoid inflation bias by believing that the continuous exploitation of expectations formed in the current period (but reflecting the response to central bank measures taken in the previous period) is not beneficial, and that the goal of the central bank will be better achieved if the central bank ignores the expectations, which are temporary anyway.¹² In other words, if the central bank applies a rule, it is important to believe, after behaving differently in the past, that expectations are never immediately aligned with actual conditions. A lag is required to align the expectations to the rule applied. Therefore, an independent central bank must implement the rule consistently, considering that after several periods, expectations will align with actual conditions pursuant to the assumption of rational expectations. Operationally, emphasizing such central bank measures must refer to the long-term characteristics of each policy procedure.

Similarly, studies on the contracts between the government and central bank have concluded that if the government provides the contract (incentives) to the central bank, where remuneration depends upon attaining the inflation target, optimal central bank performance may be achieved because it is achieving the solution based on a rule. Nevertheless, McCallum deemed this to be the second fallacy of studies on independence considering that several aspects relating to the study outcomes fundamentally fail to ignore the motivation for inconsistency, only linking that problem to the contemporaneous presence of the government and central bank.

Specifically, there is an alternative case, where the government could stipulate in the contract that the government will reduce the central bank's budget if inflation becomes too high, for example. Under such conditions, as it is relatively difficult for the central bank to avoid inflationary measures, the government also has an incentive to avoid this alternative. In other words, if the lack of pre-commitment is considered a serious problem, this should apply to an institution that encompasses the contemporaneous presence of the government and central bank. Furthermore, the problem faced will fundamentally not be overcome (analytically) by merely recommending a specification for the central bank's objective function to the constitutional phase of the political process. The problem requires

¹¹The literature (Alesina & Tabellini, 1987; Chari et al., 1989; Debelle & Fischer, 1995) states that "there is no pre-commitment technology available to the unconstrained central bank." This is due to the view that central bank behavior is in line with discretion.

¹²Expectations relate to output formation behavior (supply function), which mirrors the expectational Phillips Curve.

enforcement, thus the constitution must be effective for the government and central bank.

11.4.2.2. *Ex Post* and *Ex Ante* Accountability: The Difference between Accountability and Transparency. From the perspective of policy implementation timing, central bank accountability can fundamentally be debated in two forms, namely *ex post* and *ex ante* accountability (Bofinger, 2001). *Ex post* accountability implies that the central bank must provide *ex post* justification if the policy target is not achieved. Meanwhile, *ex ante* accountability implies that the central bank must justify deviations in the forecast from the target to be achieved. To that end, the central bank must be forward looking.

In terms of *ex post* accountability, a scenario of successful monetary policy implementation in response to demand and supply-side shocks is illustrated in Fig. 11.5.

We can now see that (*ex post*) accountability implies that the central bank must justify why it has been unable to avoid the adverse aspects, namely:

- (1) positive/negative demand-side shocks relating to over/undershooting of the inflation target; and
- (2) positive/negative supply-side shocks while achieving price stability but relating to a positive/negative output gap.

Regarding supply-side shocks, in terms of over/undershooting the inflation target and a negative (positive) output gap, there is an assumption that the central

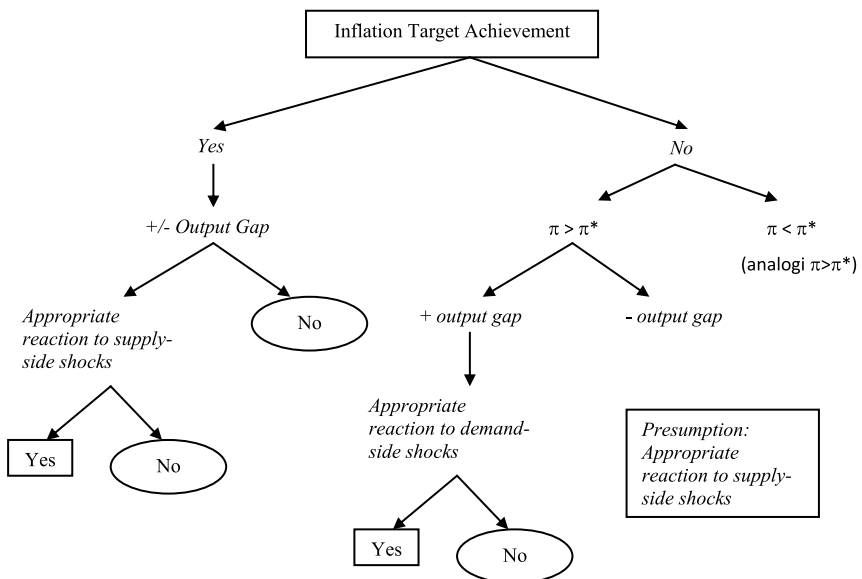


Fig. 11.5: *Ex Post* Accountability and Central Bank Reaction.
Source: Bofinger (2001).

bank reacts inappropriately but the assessment depends on the size of the output gap and inflation deviation from its target. It is important to add that of the reactions available to the central bank, the “Yes” reaction displayed in the box represents a monetary policy reaction considered successful.

A similar analysis was performed for *ex ante* accountability but the indicators used in this case were the inflation forecast and output gap. Notwithstanding, as the indicator represents a forecasted value, the analysis is more complicated. Moreover, considering the information contained in the forecast may also be provided by parties external to the central bank, if the public considers the central bank lacking objectivity when announcing the forecast, the public may instead trust the forecasts provided by external parties. Such public distrust in the central bank’s ability to provide accurate forecasts directly relates to another institutional aspect of the central bank, namely transparency.

Is the view that accountability may be ventured through transparency valid? Castellani (2002) showed that, fundamentally, there is a difference between accountability and transparency. Despite evidence that transparency plays a role in the accountability problem, the traditional “accountability through transparency” approach is considered incomplete. Several approaches, which focus on transparency as a precondition for accountable monetary policy implementation, are generally based on formal accountability concepts. The fact of the matter is that such approaches tend to fail in terms of clearly differentiating accountability and transparency.

In his study, Castellani clarified the difference, namely that accountability relates more to policy implementation (deeds), while transparency relates more to the communication strategy adopted by the central bank (words). In other words, accountability can be represented as *ex post* political intervention relating to government observations of monetary policy implementation. Meanwhile, transparency can be described as the *ex ante* decisions of the central bank in relation to the communication strategy. This is congruent with the view of Duisenberg (1999), namely that the publication of forecasts does not contribute to increase central bank accountability. In this case, central bank performance in terms of controlling inflation in the medium term is the main factor considered by the public when assessing the success, and therefore accountability, of the central bank.

11.4.3. Policy Independence and Credibility during a Crisis Period

The independence and credibility enjoyed by a central bank EBB and flow in line with the dips and gyrations of economic and non-economic factors. ITF implementation at the beginning of the 1990s was considered an underlying factor of increasing central bank independence and policy credibility in numerous countries. Meanwhile, the GFC forced many central banks to intervene as lender of last resort and institute policy to maintain financial stability. That broader mandate demanded close policy coordination between the central bank and other policy authorities, especially the fiscal authority. In addition to the benefits of harmonious policy coordination, several observers have opined that the situation precipitated a policymaking compromise, which influenced the independence and

policy credibility of the central bank and could also spur a more serious issue if the coordination hampered mandate implementation by the central bank to credibly achieve low inflation.

Would intense coordination or cooperation between the monetary and fiscal policy authorities trigger a compromise and, therefore, undermine central bank independence and policy credibility? Under normal conditions, there is a clear separation between the domains of central bank monetary policy and government fiscal policy. Nevertheless, the opposite can occur under crisis conditions.

According to Blinder (2012), this can occur because of various changes under a crisis situation. *First* is the time horizon. Under crisis conditions, when the central bank is faced with rapidly changing financial sector dynamics, the time horizon shrinks from the medium-long term to align with the shorter time horizon faced by the government. Consequently, such conditions have a propitious impact as cooperation becomes more natural.

Second relates to a shift in the objectives of the central bank, namely that it is no longer (just) combatting inflation because that could be counterproductive if the economy continues to slump; but to be jointly responsible for maintaining financial system stability. A combination of urgency and the different interests between the central bank and government requires coordination even though the central bank and government have strong differences in terms of process implementation, capabilities, legal aspects, and institutional arrangements. With the comparative advantages enjoyed by each respective institution, policy coordination between the two could better overcome the economic problems.

Third, policy coordination is critical to restore order to financial markets blighted by panic, not only operationally but also politically and psychologically. Although the central bank has the means and capability to implement policy, government support is required to ensure political legitimacy. Harmonious policy coordination between the central bank and government, and possibly other relevant policy authorities, would not only pacify market psychology but also the legislature.

Fourth, policy coordination is critical when deciding which troubled financial institutions to resolve; namely which financial institutions should be allowed to survive through bail-out funds and which should be left to close. In this case, political legitimacy in the joint decision-making process is vital. If the problem is politicized, there is a distinct possibility that the central bank or government would have to be available to take the unenviable consequences. Therefore, the different comparative advantages of the central bank and government must be exploited optimally.

Finally, per the worst-case scenario, under serious crisis conditions that threaten the solvency of the central bank's balance sheet, with potentially massive financial losses expected, the government is expected to stand firm in its backing of the central bank.

We can, therefore, conclude that, under crisis conditions, central bank policy independence could be interpreted narrowly as impossible and undesirable. Consequently, independence must encompass the broader interests through policy coordination. Nonetheless, after the crisis period has been overcome, the central

bank must prudently extricate itself from the policy compromise with the government and gradually rebuild its independence, the pace of which shall depend on the prevailing circumstances. If financial system stability has been substantially restored, the central bank must reorient monetary policy from financial system stability toward policy independence and credibility.

Therefore, similar to policy independence, policy credibility must also encompass the broader interests. In this case, anti-inflation policy credibility is (only) one aspect of credibility. Meanwhile, commitment to undertake whatever is necessary to overcome a financial crisis (as advocated by Ben Bernanke, 2008), and then meeting that commitment, is also important. Consequently, a central bank equipped to handle a financial crisis must understand that success, although not explicitly inflation control, is achieved in terms of building future policy credibility.

Considering that policy credibility is built through consistency between what is said and what is done, financial crisis resolution demands not only smart policy actions but also a high level of transparency and broad communication policy because the public, markets, and politicians have the right to a coherent explanation of the unconventional or untested policy measures adopted.

11.5. Bank Indonesia's Independence and Accountability

11.5.1. Bank Indonesia's Independence

The concept of central bank independence has been widely discussed since the establishment of Bank Indonesia as the central bank of the Republic of Indonesia in the 1950s. Mr Sjafruddin Prawiranegara, the first Governor of Bank Indonesia, hinted at potential disruptions to independence through formation of a monetary board. He stated:

it is precisely because of the nature of the work of circulation banks that their leadership cannot be influenced by the whims of political interests, therefore the government should not be given absolute power over the circulation bank. The danger is that the circulation bank might be used for the interests of political parties, which have power over the state...

With the introduction of the Bank Indonesia Act (No. 23) of 1999, which has been amended by Act No. 3 of 2004, Bank Indonesia is the central bank of the Republic of Indonesia, which has the status of an independent state institution. Bank Indonesia independence is based on the three aspects of central bank independence put forward by Bofinger (2001) as follows:

(1) Goal Independence

Functionally, Bank Indonesia has the right to decide all aspects of monetary policy and price stability in accordance with prevailing laws. Nonetheless, Bank Indonesia is not goal independent because Bank Indonesia itself

does not set the targets to be achieved, the goals are clearly stipulated in law, namely to achieve and maintain rupiah stability. In addition, it is the government that sets the inflation target, after consultation with Bank Indonesia.¹³ The lack of goal independence at Bank Indonesia is consistent with the prevailing trend of central bank independence around the world, although the degree of independence is different.

(2) Instrument Independence

Bank Indonesia has adequate jurisdiction/authority to achieve the operational target per prevailing laws. To that end, Bank Indonesia has control over the instruments used to influence inflation, while paying due consideration to the impacts on the economy in general, as stipulated in prevailing laws. Furthermore, Bank Indonesia is also proscribed from disbursing loans to the government.

(3) Personal Independence

Pursuant to prevailing laws, third parties are not allowed to interfere with task implementation at Bank Indonesia (BOG), and Bank Indonesia (BOG) must reject or disregard all forms of third-party interference. In addition, the Bank Indonesia BOG cannot be dismissed, except in cases of resignation, prolonged absence, or unlawful acts. The BOG (and/or Bank Indonesia Officers) is also immune to prosecution due to the decisions taken or policies implemented in good faith consistent with their duties and authority. The tenure of the members of the BOG is five years, with reappointment possible. The BOG is proposed and appointed by the President after approval from the People's Representative Council.

11.5.2. Bank Indonesia's Accountability

The Bank Indonesia Act (No. 23) of 1999, amended by Act No. 3 of 2004, also requires accountability in terms of task implementation, jurisdiction, and budget. The stipulation for Bank Indonesia accountability intends to ensure all stakeholders can oversee all policy measures implemented by Bank Indonesia. In addition, accountability can also be used by the People's Representative Council to appraise Bank Indonesia performance.

The principles of accountability are stipulated in accordance with prevailing laws in terms of Bank Indonesia task implementation and jurisdiction through direct public discourse in the mass media at the beginning of each year concerning an assessment of monetary policy implementation in the previous year as well as the planned monetary policy direction and monetary policy targets for the upcoming year. That information is submitted in writing to the President and People's Representative Council. Furthermore, Bank Indonesia is also required to report to the People's Representative Council on a quarterly basis or as required in line with the oversight function of the People's Representative Council.

¹³Prior to the amendment, the Bank Indonesia Act (No. 23) of 1999 stipulated that Bank Indonesia would set the inflation target. With the amendment in 2004, however, the government became responsible for setting the inflation target after coordination with Bank Indonesia.

In terms of the budget, before the fiscal year begins, Bank Indonesia is required to submit its annual budget to the People's Representative Council and government, accompanied by an evaluation of budget implementation in the current year. The operating budget of Bank Indonesia requires People's Representative Council approval through a simulated policy budget submitted to the People's Representative Council as information. Moreover, the Annual Financial Statements of Bank Indonesia is also submitted to the Audit Board of the Republic of Indonesia for auditing and the resultant opinion is published through the mass media. Bank Indonesia is also required to prepare a weekly balance sheet to be published in the State Gazette of the Republic of Indonesia.

Accountability and transparency are closely related. The central bank must be held accountable to its stakeholders for its task implementation. A more transparent institution will increase central bank accountability and, ultimately, improve central bank performance (Poole, 2001). Therefore, to increase accountability and transparency, Bank Indonesia routinely publishes various reports and publications, including the *Weekly Report*, *Monthly Economic and Financial Statistics of Indonesia*, *Monthly Monetary Policy Review*, *Quarterly Monetary and Economic Developments*, *Quarterly Monetary Policy Report*, and the *Annual Financial Statements*.

11.6. Concluding Remarks

In this chapter, we have discussed two salient institutional aspects of the central bank, namely independence and accountability, as integral parts of strengthening monetary policy credibility to achieve price stability. Chapter 10 explained that time-consistent and rule-based policy responses are crucial to build central bank policy credibility. Central bank independence, as described in this chapter, represents a form of learning the authority to achieve the price structure mandate of a competent central bank that can implement its duties and guarantee the partnership between policy and setting the rules. Meanwhile, accountability is the logical consequence of independence in pursuance of the mandate. In other words, policy consistency, including rule-based policy, independence, and accountability are all vital to build central bank policy credibility.

Relating to the principles of independence, providing a mandate of independence to the central bank, accompanied by accountability, is required ensure the effectiveness of policies to achieve the final target of professional integrity. Referring to the experiences of various countries and the underdeveloped political conditions in developing countries (typically applying a multi-party political system), central bank independence is necessary to subdue the interests of the elite that are counterproductive to the attainment of sustainable economic growth.

Although the legal facts show that central bank independence has been set proportionally in prevailing laws, implementation must be observed in the context of integrated policy coordination. This implies that the central bank's policy cannot run by itself. On one hand, central bank policy is determined as the manifestation of autonomous professionalism. On the other hand, however, the institutional and political constellation, which is oriented toward improving public

welfare, requires that the central bank not be sterile to the political process altogether. To that end, central bank policy implementation must be strengthened by harmonious policy coordination between the central bank and government or other relevant stakeholders.

The critical importance of policy coordination becomes even more apparent during crisis conditions. Numerous aspects change and problems emerge during a crisis, therefore central bank policy independence could be interpreted narrowly as impossible and undesirable. Consequently, independence must encompass the broader interests through policy coordination between the central bank and government to jointly overcome the crisis. Nonetheless, after the crisis period has been overcome, the central bank must prudently extricate itself from the policy compromise with the government and gradually rebuild its independence.

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Chapter 12

Policy Transparency and Communication Strategy

12.1. Introduction

Transparency and communication are integral parts of strengthening monetary policy credibility and central bank policy in general. How the statements of the chairman of the US Federal Reserve are always anticipated and influence developments on the US and global financial markets and economies is a solid example. Over the past decade, central banks around the world have highlighted transparency by improving their communication strategies and providing more information to the public. Senior central bank officers routinely clarify the monetary policies taken to the public. Furthermore, most central banks publish the inflation and economic forecasts underlying their monetary policy orientation. In fact, many central banks clarify how the monetary policy framework is applied, including publishing the projection models and results of other research.

Notwithstanding, there remains an ongoing debate in theory and in practice over how transparent a central bank should be. Theoretically, economists tend to suggest that there is never too much information given to the public (Blinder, Goodhart, Hildebrand, Lipton, & Wyplosz, 2001). Since the emergence of classical economics, imperfect information has long been viewed as a source of market imbalances that create distortion in the allocation of economic resources, which translates into suboptimal public welfare.¹ Likewise, in the context of the debate between classical and Keynesian economics with rational expectations theory, although surprise or unanticipated monetary policies influence the real economy, the central bank's desire to manipulate expectations will not be optimal for the general economy (Barro, 1977; Cukierman & Meltzer, 1986; Lucas, 1972; Sargent, 1973; Sargent & Wallace, 1975). In a broader context, demand for monetary policy transparency as a public policy is the embodiment of democratic accountability (Blinder et al., 2001). In the words of Milton Friedman, as cited

¹Refer, for example, to the market for lemons theory developed by Akerlof (1963), which analyzes market imperfections due to asymmetric information.

by Faust and Svensson (2000), for instance, the reluctance of the central bank to communicate monetary policy is “avoiding accountability on the one hand and achieving public prestige on the other.”

The policymakers, including the central bank, view information differently. In the past and at some central banks even today, secrecy remains a key concern, with the central bank restricting external and internal communications. In general, the rationale and framework underlying monetary policy is the preserve of the most senior leadership of the central bank. Such information is viewed as a recipe or instrument private to the central bank because it determines monetary policy effectiveness in terms of influencing the economy. Concerning the theory of asymmetric information, this view can be explained by the superiority of the information held by the central bank as the most important element of monetary policy effectiveness. Similarly, information confidentiality also supports monetary policy effectiveness because only surprise/unanticipated monetary policy has an effect on the economy. Institutionally, many central banks in the past were not independent, thus demand for transparency as a key to fulfilling monetary policy accountability was also not forthcoming.

Nevertheless, it seems that that era has changed. Convergence between central bank policy transparency theory and practice, particularly monetary policy, has already begun, especially as more and more central banks expand transparency and their monetary policy communication strategy. This chapter specifically focuses on that problem over six sections. After the introduction, the second section puts forward several concepts and empirical dimensions underlying the importance of transparency and its implications on policy effectiveness and credibility. The third section explores the theoretical models developed that underlie policy transparency at numerous central banks. Thereafter, the practices and empirical studies on policy transparency and communication strategies in several countries will be presented. Then, before the concluding remarks, Bank Indonesia's efforts to increase policy transparency and improve the communication strategy are presented.

12.2. Conceptual Dimension of Policy Transparency and Communication Strategies

There are at least three factors that can explain the phenomenon of increasing monetary policy transparency and improving communication strategies in many countries (Amato, Morris, & Shin, 2003). *First*, greater central bank independence demands increased accountability. In general, increasing accountability translates into more central bank communication concerning the direction of monetary policy operations and the confidence therein. Only through increasing transparency can a fair appraisal of central bank performance be conducted. Therefore, regulating the relationship between an independent, transparent and accountable central bank and the government and public is becoming more important. Establishment of the European Central Bank (ECB), which avoided parliament and government interference, is a good example.

Second, more and more central banks in advanced countries and emerging market economies (EMEs) are implementing the Inflation Targeting Framework (ITF) as their monetary policy framework. As discussed in Chapter 8, although the regimes applied by central banks vary, all central banks that formally implement ITF stress the importance of communicating monetary policy to the public. Through regular public communication, the monetary policies instituted by the central bank to achieve the inflation target will be better understood by the public. This is important not only to increase central bank credibility and accountability but also to anchor public inflation expectations to the desired inflation target.

Third, financial markets are becoming more developed in many countries. Transactional behavior and price formation in financial markets are predisposed to market expectations concerning the economic outlook, including inflation and the direction of the central bank's policy rate. The discussion on monetary policy transmission mechanisms presented in Chapter 5 evidenced the importance of the expectations transmission channel. Furthermore, monetary policy influences transactions and financial asset prices through other transmission channels, such as the interest rate, money supply, credit, exchange rate, and asset price channels. Managing and manipulating expectations in financial markets, therefore, are important parts of monetary policy, which will clearly not be achieved without transparency and a sound communication strategy by the central bank.

As a contemporary issue of monetary economics, various viewpoints have been developed on what, why, the extent and how monetary policy transparency and communication is undertaken by a central bank. Nevertheless, the differing views fail to reach the same conclusion. Consequently, it is worth reviewing the different opinions contained in the literature before discussing transparency practices at central banks and their contribution to economic performance in more depth.

12.2.1. Several Viewpoints Concerning Transparency

In general, Poole (2003) defined monetary policy transparency as “accurately conveying accurate information, including all the information market participants need to form opinions on monetary policy that are as complete as possible.” In the context of the Code of Good Practices in Monetary and Financial Policies developed by the IMF, Sundarajan, Das, and Yossifov (2003) provided more concrete understanding that

monetary and financial transparencies refer to an environment in which the objectives of the policy; its legal, institutional and economic framework; policy decisions and their rationale; data and information related to monetary and financial policies; and the accountability of the policy making body are provided to the public in an understandable, accessible and timely way.

This is consistent with the views of Geraats (2001), who placed transparency in the phases of providing information on monetary policy to the public.

Geraats (2001) separated transparency into five aspects, namely: (1) disclosure of the final policy goal, such as the explicit inflation target (political transparency); (2) disclosure of the economic data, models, and forecasts delivered by the central bank (economic transparency); (3) information relating to the monetary policy strategy and the associated internal considerations, for example, by publishing the minutes and voting records (procedural transparency); (4) communication of the policymaking process, for example, a change in the policy rate and statements regarding the future policy direction (policy transparency); and (5) disclosure of policy implementation, market interventions, and control errors (operational transparency or commonly referred to as market transparency).

Differentiating transparency facilitates a review of how the views have developed in the literature in order to understand the motives and effectiveness of transparency itself. Political transparency has been analyzed by Faust and Svensson (2001) as well as Nolan and Schaling (1996). The analyses showed that clarification regarding the central bank's preferences and desired goals of monetary policy had the propitious impact of reducing inflation bias as well as fluctuations in inflation and job opportunities. This is because such assurance informs the public of the central bank's preferences in terms of the ultimate goal of monetary policy, in this case inflation, and, therefore, reduces the disparity between the rate of inflation desired by the central bank and the inflation expectations of the public.

Meanwhile, procedural and policy transparency, specifically in terms of publishing the minutes, voting records and policy directives, has been discussed by Goodfriend (1986), Buitier (1999), and Issing (1999). Formally, the publication of individual voting records was analyzed by Gersbach and Hahn (2000). Regarding policy transparency, in particular clarifying or publishing the direction and decisions of monetary policy, most economists view the positive contribution on expectation formation by economic agents and the effectiveness of monetary policy itself. Nonetheless, concerning procedural transparency, specifically the publication of minutes and voting records, there remains a difference of opinion between economists and central bankers. Hitherto, only a few central banks have published the minutes of their meetings and voting records of the monetary policymakers.

Several models of operational transparency have also been developed. In their seminal paper, Cukierman and Meltzer (1986) proposed operational ambiguity in terms of monetary policy to improve controlling errors when striving to achieve the goals set. Meanwhile, Faust and Svensson (2000, 2001) provided a theoretical foundation for the differences between the inability to control monetary policy and operational transparency, while also showing that transparency could ameliorate public welfare. In a similar model to Faust and Svensson (2001), using New Keynesian economics rather than the standard Phillips curve, Jenson (2002) found that increasing transparency could undermine public welfare and, therefore, advocated confidentiality in terms of monetary policy.

Economic transparency has also been discussed in relation to several models based on central bank publications of economic and inflation forecasts as well as assessments of public expectations and the formal models used.

Gersbach (1998), Cukierman (1999), and Geraats (2001) argued that publishing the central bank's projections eliminates asymmetric information concerning shocks in the economy. Notwithstanding, differences in the assumptions regarding monetary policy transmission channels and information on the central bank's preferences have produced disparate conclusions, where Gersbach (1998) and Cukierman (1999) showed how transparency undermined public welfare but Geraats (2001) showed how transparency improved public welfare. On the other hand, Tarkka and Mayes (1999) argued that not only do the public not know the central bank's preferences but also do not have access to the central bank's assessment of public inflation expectations. Therefore, by publishing its forecasts, the central bank could eliminate such public uncertainty and, hence, improve monetary policy effectiveness moving forward.

12.2.1.1. Scope of Transparency: Full Versus Limited. From an economic perspective, the main argument supporting the need for full transparency is that monetary policy would be most effective if correctly anticipated by the markets. That argument is based on the three characteristics of effective monetary policy signaling (Blinder et al., 2001). *First*, monetary policy generally works through variables that are moved by market expectations. Consistent with Blinder (1998), short-term interest rates are the most controllable variable by the central bank in many countries and also have a broad effect on the economy. Meanwhile, long-term interest rates, asset prices and exchange rates, despite also having a strong and wide-ranging impact on the economy, are difficult to control by the central bank. The relationships between various financial variables and short-term interest rates involve the expectations formed in the markets. Less volatility in the expectations leads to a more stable and easier-to-predict relationship between monetary policy and its impact on the economy. To some extent, market expectations are influenced by the future direction of central bank actions, therefore, monetary policy would be more effective if easily and accurately anticipated by the markets.

Second, the transmission channel from the interest rate to the real economy and inflation involves price and wage formation at the producer and consumer levels. Part of the influence of monetary policy lies in its impact on economic dynamics, through the traditional backward-looking Phillips curve, as explained by Gordon (1998), and partly through expectations in the forward-looking Phillips curve channel, as found by Clarida, Gali, and Gertler (1998). The first channel typically operates more slowly and disruptively, while the second channel is faster yet more difficult to predict. Consequently, transparency would improve credibility and credibility would strengthen the second channel and, therefore, improve the effectiveness, predictability, and speed of monetary policy transmission. In fact, price and wage formation would adjust quickly if what the central bank is doing and trying to achieve are known.

Third, transparency also reduces the costs incurred by changing policy direction. If economic dynamics change or differ from what was anticipated, the direction of monetary policy may need to be adjusted. In this regard, central bank concerns that a policy change would confuse the public and undermine credibility are understandable. On the other hand, however, the original policy

direction could delay the policy adjustment and lead to suboptimal policy. Therefore, it is fundamental for the public to fully comprehend the policy regime adopted so that a policy change would be easier to accept and could reduce the costs incurred.

Based on those three arguments, Blinder et al. (2001) concluded that the basis for monetary policy transparency is incredibly robust. Fundamentally, the rationale lies in the phenomenon of asymmetric information, namely that the central bank has greater knowledge about itself, its instruments and desires compared to the public. Such conditions create a misunderstanding that undermines monetary policy effectiveness as described above. Although the markets will constantly strive to understand central bank behavior, if the central bank fails to clarify its knowledge of the economic situation and monetary policy direction pursued, misunderstandings will continue to occur. It would be better for the central bank to transparently select the regime and procedures of monetary policy implementation. In other words, this view recommends, as a general rule, that all aspects must be communicated publically, except in special cases where confidentiality is necessary.

Considering the clear and fundamental need for broad transparency, why did central banks limit their communication with the public for so long? Why did central banks in the past keep the considerations underlying monetary policy a secret or prefer creative ambiguity? A number of considerations can explain such developments.

First, only unanticipated monetary policy is effective in terms of influencing the economy. The seminal argument for limited transparency was proposed by Cukierman and Meltzer (1986) by assuming that, in an economy with rational players, the central bank's actions would have already been considered when forming the prices of goods, wages, and interest rates in the markets. Under such conditions, monetary policy that has been anticipated would not effectively influence the economy and lead to the short-run neutrality of money. Consequently, to successfully steer economic developments in the desired direction, the central bank must limit transparency in order to act as leader in this game, implying that the central bank must formulate monetary policy that is full of surprises. According to Blinder et al. (2001), this argument, although common in economic theory, lacks the support of convincing empirical evidence. Furthermore, there is also a lack of cogent reports that central banks try to create surprises when implementing monetary policy.

Second, limited transparency improves monetary policy's ability to influence economic growth and job opportunities in the near term. This argument is based on the view of inflationary bias proposed by Kydland and Prescott (1977), namely that if the central bank wishes to stimulate economic growth, or reduce unemployment, inflation surprises must be created from time to time. This ability would be lost, however, if monetary policy was implemented transparently. Of course, the need for transparency would fade if the central bank already enjoyed a high level of credibility. With a high level of central bank credibility, the public would continue to trust the actions of the central bank and, therefore, the central bank would have more space to create inflation surprises and influence output.

According to Blinder et al. (2001), there is no firm evidence concerning inflationary bias from central banks in advanced countries. Despite several cases of inflationary policies implemented by central banks to stimulate short-term output, such moves typically stem from past political pressure and usually fail to hit the target. With the growing trend of central bank independence around the world, such arguments lack a solid foundation.

Third, however broad the respective central bank's scope of transparency is, the markets and public may still not be able to accurately capture the corresponding monetary policy signals and, therefore, asymmetric information would still persist. In fact, for the skeptics and speculative investors, such information from the central bank could be manipulated to create inaccurate and exaggerated expectations in pursuit of profit. Financial market volatility could increase, coupled with panic and herding behavior, ultimately holding the central bank hostage to market sentiment. According to Blinder et al. (2001), that argument is baseless, showing that the typical sources of market confusion are insufficient and inaccurate information from the central bank. Fundamentally, disclosure would eliminate misunderstanding, the cost of which, for monetary policy effectiveness, is prohibitive. A better solution would be to provide clear and complete information concerning the future direction of monetary policy, such as the direction of the interest rate, so that the markets, especially the bond market, can formulate accurate expectations of the interest rate term structure in the markets. In the words of Tarkka and Mayes (1999) and Winkler (2000), full public transparency regarding the central bank's preferences would precipitate convergence between the heterogeneous views found in the financial markets and accurate expectations.

12.2.2. Transparency, Monetary Policy Regime, and Democratic Accountability

The various opinions among economists described above demonstrate the different considerations underlying the rationale of monetary policy transparency. Likewise, in practice at many central banks, the basic considerations and communication strategies differ. On one extreme, communication could merely be limited to specific aspects of monetary policy. On the other extreme, however, communication could be oriented toward achieving the maximum degree of transparency concerning various aspects of monetary policy.

In this respect, it would be interesting to discuss the linkages between transparency and the monetary policy regime applied by the central bank, as described by Blinder et al. (2001). To that end, a central bank could be considered transparent if it provided adequate information to the public at all times to understand the monetary policy regime, thereby facilitating cross-checks and providing a clear assessment of whether the actions of the central bank were consistent with the regime applied. Theoretically and as practiced by central banks, the monetary policy regime could vary from rules to discretion-based. With such understanding, communication seeks to provide various detailed information on the procedures, data, policy formulation process, and so on, as applied by the central bank.

The currency board regime with a simple automated rule occupies one extreme of policy transparency. Under such a regime, monetary policy is based on an exchange rate peg with a simple rule for the central bank to provide the exact same amount of money supply as the change in reserve assets required to support the fixed exchange rate. As the regime converts to an exchange rate peg, simple transparency in the form of an exchange rate rule is maintained but the central bank enjoys additional monetary policy flexibility due to the possibility of instituting devaluation policy or revaluating the exchange rate, while being exempt from providing the same amount of money supply as the change in reserve assets.

A crawling peg or managed floating regime affords greater flexibility because the exchange rate is allowed to fluctuate within a given range, in addition to the advantages offered by the exchange rate peg mentioned previously. Therefore, monetary policy transparency depends on the width of the exchange rate range (upper and lower bounds). Meanwhile, clarification regarding the discretion-based foreign exchange intervention conducted by the central bank is not usually published. In the context of monetary policy, flexibility is required to increase central bank autonomy in terms of orienting monetary policy toward the domestic economic goals, particularly inflation, away from external influences.

With a more flexible exchange rate system, the monetary policy regime is more often oriented toward the domestic economic goals. As explored in Chapter 8, an inflation targeting regime, in this case the ITF, entails transparency at least in terms of the ultimate goal and instruments of monetary policy. The final goal of monetary policy is price stability, while also determining and announcing the medium-long term inflation target. Meanwhile, monetary instrument transparency is reflected in the interest rate direction and operational target in the current period to achieve the future inflation target. In practice, flexibility per this regime generally lies in the need to publish forward-looking economic and inflation forecasts, the models used as well as other types of operational transparency.

A monetary policy regime based more on discretion than a rule involves more complex transparency. This is usually applied by central banks with multiple mandates, not only to achieve price (inflation) stability but also to stimulate economic growth and create job opportunities for instance. Furthermore, the complexity of the transparency not only lies in the lack of clarity concerning the final target, whether it be inflation or economic growth or something else, but also in the linkages between the strategy and operational target set by the central bank. In general, central bank communication emphasizes qualitative clarification regarding the future direction of the economy and inflation based on the current operational target of monetary policy. The Federal Reserve, for instance, generally announces its decisions and the desired Federal Funds Rate (FFR) based on qualitative assessments of the economic and inflation outlooks, without providing a detailed explanation concerning the formal framework used to formulate monetary policy.

12.2.2.1. Transparency and Democratic Accountability. The trend of democratization that has affected many countries, particularly in the past decade, has

increased calls for transparent public policies, including monetary policy. The central bank is a public institution and, therefore, must be held accountable for all actions and policies implemented. The principles of accountability, therefore, are the basis of the requirement for information on the monetary policy implemented by the central bank to be published, unless there is a strong reason not to. That exemption could be justified if the information submitted periodically by the banking industry and financial institutions to the central bank, along with the variety of other sensitive information, could undermine the integrity of financial market operations. Beyond that, other information is required by the public as the manifestation of democratic accountability (Blinder et al., 2001).

The application of democratic accountability rests on the central role of the communication strategy to increase central bank accountability. Whichever legal mechanisms must be followed to fulfill monetary policy accountability, the central bank will never be able to soundly implement its duties without the backing of the stakeholders. Under certain circumstances, the central bank must implement unpopular policies to achieve the goals, such as price stability. For example, the monetary tightening required to control inflationary pressures would raise the interest rate, which could have an adverse impact on short-term economic growth. Pursuing such a policy would fail to garner public support if trust in the central bank, including the monetary policy direction, targets and commitments, has not been built. In other words, monetary policy effectiveness can only be improved if the motives and procedures of the central bank are clearly understood by the public. The application of democratic accountability principles emphasizes the importance of a central bank communicating constantly and transparently, during the good times and bad, with the relevant stakeholders to clarify the monetary policy stance.

In practice, there are variations in the application of democratic accountability principles in terms of central bank monetary policy transparency from one country to the next, depending on the democratic progress achieved and prevailing social norms. In several countries, as discussed in Chapter 11, the process of democratization has been accompanied by broader central bank independence. In terms of the regulatory, intuitional and monetary policy frameworks, greater independence has led to a more obligations in terms of transparency and accountability. This is congruous with the proliferation of the ITF in many countries. In this case, clearer transparency guidelines regarding the price stability goal of monetary policy, announcing the inflation target, clarifying the monetary policy framework, publishing economic and inflation forecasts as well as announcing the direction and operational target of monetary policy are integral parts of ITF implementation.²

In several other countries, particularly where the central bank does not enjoy independence in practice or independence is not contained in prevailing laws, the application of democratic accountability in terms of monetary policy transparency is not clearly observable. Under such conditions, the central bank remains

²Refer to the discussion on ITF implementation at various central banks in Chapter 8.

an extension of the government and, therefore, monetary policy is formulated and applied as a part of the government's economic policy. Consequently, clarity concerning the goals, targets, and monetary policy framework is not as forthcoming as from a central bank that enjoys a high degree of independence. Although some of these central banks announce their monetary operations, the linkages to the ultimate target of monetary policy, desired and prioritized, remain opaque.

12.2.3. Monetary Policy Communication Strategy

The discussion above demonstrates the strength of the argument for monetary policy transparency by the central bank, although full transparency is not possible to achieve and, therefore, market misunderstanding could possibly persist. The question is therefore, what scope of information should be communicated, by which methods and to whom?

12.2.3.1. Scope of Communication. The principle underlying monetary policy transparency is that the information communicated helps the public to understand, and anticipate, the central bank's decisions as a logical conclusion to the policy sequence oriented toward achieving the final target. Per that rationale, Blinder et al. (2001) substantively opined that the scope of information communicated to the public in accordance with the corresponding priority involves the following aspects: (1) goals; (2) methods; and (3) decision-making;

(1) Goals:

The central bank must clarify what it hopes to achieve from its monetary policy, including the final target (such as price stability) or the short-term goals (namely the operational target, such as the short-term interest rates, and intermediate targets, such as money supply, medium-term interest rates, and the exchange rate). The public understands that determining the targets represents a political decision and often entails a dilemma. The public wants to see rational targets, particularly in support for public prosperity, and whether the central bank is consistent and serious in terms of achieving the goals when implementing monetary policy.

The most fundamental aspect to be communicated is the long-term goal because this will have the biggest impact on public welfare. Therefore, the scope of information will depend on the monetary policy regime adopted. This is comparatively clear and simple for countries applying a pegged exchange rate system because the exchange rate, point or range, is the goal of monetary policy. Similarly, in ITF countries, price stability is usually stipulated in prevailing laws and the inflation target, whether it be set by the government or the central bank itself, is also announced publically.

Conditions become less clear for central banks with a more ambiguous goal, for instance "achieving price stability, while stimulating sound national economic development," such as the Bank of Japan (BOJ), or when there is a dual mandate for monetary policy, for example, "achieving price stability and creating broad job opportunities," such as the US Federal Reserve.

The short-term goals must also be communicated. Monetary policy affects inflation and economic growth after a lag, therefore information on the short-term goals should encompass the operational and intermediate targets to be achieved. Transparency, in terms of the operational target, for example, the short-term interest rate target, patently needs to be communicated because the information contained within could be utilized by the public as a measure of the monetary policy actions taken by the central bank to achieve the long-term target.

The case becomes less clear regarding the intermediate targets, not only because the degree of uncertainty is linked to the operational and final targets, but also because of the ever-present dilemma. Previously, we discussed the argument for transparency concerning the intermediate target, primarily to form market expectations and improve monetary policy efficiency. Nevertheless, such transparency, especially if the dilemma between controlling inflation and stimulating job opportunities is present, could risk public opposition and open the door to political intervention from the government or parliament.

(2) *Method:*

A difference of opinion also prevails concerning the extent to which a central bank needs to be transparent about the methods used to determine monetary policy, particularly in terms of the economic forecasts and models used. In general, the opinion to limit transparency is based on the argument that the economic models and projections use specific assumptions, contain a degree of uncertainty and could have a potential impact on central bank credibility if proven erroneous. Blinder et al. (2001) recommended the opposite by advocating transparency, particularly in terms of the key features and basic assumptions used in the economic forecasts and models, without necessarily detailing the minutia. In addition to forming expectations on the financial markets and enhancing public understanding of the central bank's monetary policy, the public would also gradually understand that the projections still contain the risk of inaccuracies.

On the other hand, the central bank is generally transparent in terms of the practical methods employed, namely the monetary operations applied to achieve the operational target. Each central bank that announces its operational target decisions, for instance the level or direction of changes in short-term interest rates, also directly announces its monetary operations, oriented to achieve the operational target through open market operations, discount facilities and other facilities. Exemptions are typically put in place regarding foreign exchange intervention by the central bank, which is usually undertaken based on certain considerations by the central bank. The importance of such procedural transparency is based on the need to avoid and minimize shocks in the markets, even within a brief period.

(3) *Decisionmaking:*

Transparency concerning the operational target decisions made by the central bank is always monitored publically, not only in terms of forming

expectations but also to check the consistency of monetary policy. In this case, central banks in several countries usually announce changes in the operational target, for instance decisions relating to short-term interest rates, immediately after the decision has been taken. The announcement is made through a press release, sometimes with an accompanying explanation and sometimes without. Such practices are becoming more common at central banks, a positive development in terms of monetary policy transparency.

Nevertheless, there remains a difference of opinion regarding the need for the central bank to announce its preliminary plans about future monetary policy. The traditional viewpoint argues that forward-looking information is still considered taboo because of concerns that such information could be misinterpreted by the markets, thus triggering excessive expectations regarding the future orientation of monetary policy.

On the other hand, to increase transparency and improve the formation of expectations, a more moderate view tends to support central bank measures to clarify the direction of monetary policy with the markets, politicians, and public. In this case, transparency involves clarifying the future direction of the economy and inflation, the concerns that need to be addressed, as well as the direction and considerations underlying monetary policymaking.

Several central banks have already begun to announce the forward direction of their monetary policy. The most advanced is the Reserve Bank of New Zealand, which has published its monetary policy contingency plan for the upcoming three years since 1997. The contingency plan emphasizes that only monetary policy decided in the current period will be implemented, while the future policy direction will depend on the unfolding dynamics. Likewise, the US Federal Reserve began clarifying its monetary policy direction in May 1999. Although qualitatively in line with confidence concerning the future direction of the economy and inflation, the move is often referred to as a preventative action. On the other hand, the Bank of England and ECB are more vague when indicating the forward direction of monetary policy.

Meanwhile, in the context of financial and monetary policy transparency, Sundarajan et al. (2003) proposed three fundamental elements of sound policy transparency, namely: (1) clearly and consistently stated policy goals, with periodic clarification regarding performance and the underlying considerations; (2) a solid foundation, legally, institutionally and economically, for the policies instituted; and (3) data and information availability to create a sound view of the monetary and financial policies implemented as well as the implications on the financial choices of individuals and companies.

The three basic elements outlined above underlie the “Code of Good Practices on Transparency in Monetary and Financial Policies,” which was developed by the IMF in 1999 and is now adhered to by many member states (IMF, 2003).

In more detail, the scope of monetary policy transparency contained in the document is as follows³:

- Clarification regarding the role, jurisdiction, and goals of the monetary policy authority, interpreting and explaining the final goal and institutional framework of monetary policy, as well as the institutional relationship between the monetary policy authority and other financial authorities.
- Disclosure of the monetary policy formulation and reporting processes, clarifying the framework, instruments, and targets set to achieve the final target of monetary policy; changes in the monetary policy instruments; as well as the progress in terms of achieving the final target of monetary policy and the outlook.
- Public availability of monetary policy information, explaining the presentation and publication of central bank data in accordance with the prevailing standards; balance sheet reporting and annual financial statements, as well as the salient economic, monetary, banking and financial sector developments.
- Accountability and integrity of the monetary authority, covering specific regulations concerning the accountability mechanism, such as periodic monetary authority clarification before parliament or the public regarding monetary policy implementation; availability of audited financial statements; internal procedures per good governance; as well as behavioral guidelines (code of ethics) and legal protection for monetary authority employees.

12.2.3.2. Communications Methods. Sundarajan et al. (2003) proposed four basic dimensions that determine the credibility of policy transparency, namely: (1) the means or media used; (2) timeliness; (3) periodicity; and (4) the quality and scope of the information submitted. On top of those four dimensions, Ehrmann and Fratzscher (2005) added another: timing. The five dimensions of transparency can be explained as follows:

- There are various means and media used in terms of monetary policy transparency, such as: (a) clarification through the publication of official documents; (b) clarification through the mass media or before parliament; (c) direct clarification with the public; and (d) other forms of clarification. The various means and media can be used simultaneously per the desires of the monetary authority to effectively expand its transparency. In many ways, transparency can be expanded through public discourse to seed holistic understanding of the monetary policy instituted by the central bank.
- Timeliness in terms of submitting information is crucial to effectively create transparency. That information must be useful and relevant in terms of monetary policy implementation by the central bank. The information should be published immediately after a decision has been taken or when the data become

³For a detailed explanation on financial policy transparency, refer to IMF (2003).

available. Delaying the publication of such information not only makes it out-dated but also contains the risk of a leak and could trigger spurious rumors about the actual monetary policy decisions and data.

- Periodicity in terms of submitting information is also vital to maintain the credibility of monetary policy transparency. The publication schedule for monetary policy decisions, once set, should be maintained and respected. For example, if the central bank has already resolved to announce the results of the decision on the operational target of monetary policy, be it short-term interest rates or money supply, on a monthly basis, then the announcements must be delivered consistently on a monthly basis. This is important to formulate market expectations and expectations among economic agents. Delaying the announcement must be avoided, especially if the news is bad, because this would not only challenge the spirit and meaning of transparency but could also undermine monetary policy credibility.
- The scope and quality of the information provided is also critical to support monetary policy transparency. The scope of information will be determined by a central bank assessment of the types of monetary policy information to be published. The emphasis of transparency must be placed on the materiality and relevance of the information presented. The goal of transparency will not be attained if contradictory information is published or if the language used is too technical or ambiguous.
- The timing of publishing information is also considered important to monetary policy effectiveness. In this respect, it is important to understand whether the publication of information under certain circumstances is required, when it is required and the intensity required, considering that the policies communicated could be understood differently by financial market players before and after a policy is announced to the public.

12.2.3.3. Communication Targets. To whom monetary policy transparency and communication is targeted is a reflection of the democratic accountability principles described previously. Blinder et al. (2003) suggested four main targets of central bank communication, namely: (i) the mass media and public; (ii) the government and parliament; (iii) the financial markets; and (iv) central bank observers. The scope of information and the communication means would, therefore, depend on the four targets.

(i) *Mass media and public:*

In general, the public does not know in detail, nor can the public anticipate, the monetary policy of the central bank. Nevertheless, the public understands the importance of price stability and will complain if inflation increases. Furthermore, the public also monitors interest rate and exchange rate developments, especially if large fluctuations occur. Ultimately, the public sees the central bank as the authority with control over economic growth and job availability.

Such societal conditions do not exclude the public from being the target of central bank communications. In fact, public sentiment has become the focus

of central bank communication programs in many countries, using the various information channels available. A range of brochures are provided to the public, whether published, explained directly or through the mass media and official website of the central bank. Furthermore, central bank officers routinely appear at press conferences, interviews and briefings to relay the performance and background behind the monetary policies implemented. In general, the public gets its information from the non-specialized mass media, including the television, radio, and newspapers.

The specific challenge facing central banks lies in the communication strategy through the specialized mass media, providing financial analyses, including the central bank's monetary policy. In this case, the specialized mass media wishes to provide more detailed commentary and analysis concerning the pros and cons of monetary policy. Nonetheless, the content of the analysis, especially on the television, is sometimes inaccurate and leads to misunderstanding. That is why it is important for the central bank to furnish reporters with the background and a greater understanding through periodic workshops, briefings and other means. Equally as important is to provide observers of central bank policy a clear perspective and analysis to help mould public opinion.

(ii) *Government and parliament:*

In terms of democratic accountability, the government and parliament represent the public but do not always have the same agenda as the monetary policy of the central bank. On one hand, the government has a keen interest in monetary policy because of its effect on several important economic indicators, such as inflation, interest rates, exchange rates, and economic growth. Nevertheless, greater government attention on populist policies sometimes places the government's focus on the near term, particularly in terms of achieving robust economic growth and creating broad job opportunities despite reducing the weight and importance of controlling inflation and the implications on the central bank's policy rate.

Under such conditions, conflicting views between the government and central bank could emerge when determining the inflation target and its implications on interest rates and economic growth. Furthermore, the proliferation of greater central bank independence has occasionally raised questions regarding the most appropriate relationship format between the government and central bank. Herein lies the importance of a close relationship and communication strategy between the central bank and government. However central bank independence is implemented, monetary policy effectiveness will be influenced by synergy with fiscal policy because both are integral elements of macroeconomic policy. Consequently, a communication strategy that strengthens monetary-fiscal policy coordination is fundamental, among others, through periodic meetings between the Minister of Finance and Governor of Bank Indonesia as well as their subordinates at both institutions to discuss macroeconomic projections as state budget assumptions as well as the various aspects of monetary, fiscal, and other macroeconomic policies based on the duties and authority of each respective institution.

Meanwhile, parliament generally represents the public in terms of monitoring and appraising central bank performance, including monetary policy implementation. To that end, the central bank is typically required to submit periodic reports to parliament. In many cases, a senior central bank officer will routinely appear before parliament and follow the dialog covering various policy aspects in the interests of the public. The quality of the dialog will depend on the support staff in parliament as working partners of the central bank. The dialog between the central bank and parliament is heavily reported in the mass media and, therefore, complete, unambiguous, and accurate clarification would support the credibility of monetary policy transparency in the eyes of parliament and the public.

(iii) *Financial markets:*

The financial markets always maintain a keen interest in monetary policy because of its strong influence on market performance. In terms of monetary policy, the financial markets are an important channel where the monetary decisions of the central bank are transmitted to the economy and, eventually, to the achievement of the final monetary policy target. Nonetheless, this channel is heavily influenced by expectations, therefore the central bank's ability to reassure the financial markets is critical. That is why the central bank always allocates a large portion of its communications to the financial markets.

Blinder et al. (2001) noted several aspects that demand attention when the central bank communicates with the financial markets. *First*, financial system stability (FSS) fundamentally underlies the central bank's attention because of its role in terms of financial intermediation for the real sector as well as monetary policy transmission. It is important to remember that a focus on FSS cannot shackle the central bank when formulating its monetary policy. In many cases, however, particularly when there is a tangible threat to FSS, it is best for the central bank to clarify the situation to the public. *Second*, it is important to realize what may be considered sound for the financial markets may not always be best for the public. Fundamentally, the financial markets like volatility, for example, interest rate or exchange rate volatility, as a source of profit, while in the interest of the public, financial stability is preferred. Moreover, the financial markets operate on a shorter horizon, from second to second even, while the public cares about the effect of monetary policy on inflation and economic growth in subsequent periods, in addition to interest rate and exchange rate developments. In this respect, monetary policy transparency can bridge the sometimes disparate interests of the financial markets and public. Besides, greater transparency feeds through to less volatility on or created by the financial markets from the formation of expectations that are misaligned with the monetary policy direction. In addition, transparency also allows observers to assess the performance and direction of monetary policy as a correction if erroneous expectations are formed on the financial markets.

(iv) *Central bank observers:*

Central bank observers encompass the financial markets, media and academia. They have deep expertise and understanding regarding the technical aspects, which allows them to interpret the central bank's monetary policy. Therefore, such observers play an important role in terms of clarifying the actions of the central bank and disseminating the various information communicated by the central bank to the public. In fact, some of the observers will have the ability to forecast the direction of monetary policy, thus playing an important role in terms of transmitting monetary policy to the financial markets or forming public opinion in general. Such abilities, given adequate information through an appropriate communication strategy, will play a strategic role in determining the effectiveness of the central bank's monetary policy.

For the central bank, such observers could be considered a friend or a foe. The key lies in how far the central bank can communicate and persuade the observers regarding the rationale and orientation of monetary policy implementation. A high level of transparency would reduce the risk of misinterpretation by a section of the observers, thus supporting the clarification and dissemination of information by the central bank. Nonetheless, providing too much information, especially confidential information, is not always beneficial because the risk of misusing that information to form public opinion could paint the central bank into a corner. Therefore, the most appropriate strategy would be to provide as deep of an explanation as possible concerning the aspects of monetary policy that should be in the public domain and select more limited information regarding the more confidential aspects.

12.3. Theoretical Models on Policy Transparency: Conservative Central Bank and Imperfect Transparency

This section shows the possible effects of imperfect transparency on economic dynamics. Conceptually, the problem of imperfect transparency could occur when the central bank has certain information on the characteristics of a shock and the policy response to influence the economy is not known by the public (Cukierman, 1992, 2000); or when the central bank ambiguously states its purpose (Cukierman & Meltzer, 1986); or when the public faces uncertainty regarding the central bank's preferences (Nolan & Schaling, 1996). To accommodate such possibilities, when developing their theoretical model, Demertzis and Hughes Hallett (2003) observed imperfect transparency from two aspects, namely: (1) political imperfect transparency concerning the policy goals and priorities; and (2) economic imperfect transparency concerning the information relating to policy-making, shocks, target values, and so on. For both respective conditions, imperfect transparency triggers disruptions that impair the correct formation of public expectations, thus influencing more general economic activity.

Fundamentally, the theoretical model of transparency developed was adopted from the basic model developed by Rogoff to explain the issue of a conservative central bank (Rogoff, 1985), as explored in the previous chapter. In this case, the central bank's loss function and supply constraint function, à la Lucas, were formulated as follows:

$$L = \frac{1}{2} E[\pi^2 + b(y - k)^2] \quad (1)$$

$$y = \pi - \pi^e + \varepsilon \quad (2)$$

where y and π , respectively, are output and inflation expressed in terms of deviation from their steady state value. Meanwhile, $b > 0$ is the output stabilization weight. The solution for the policy variables of the model are as follows:

$$\pi = bk - \frac{b}{1+b} \varepsilon \quad (3a)$$

$$\pi^e = bk \quad (3b)$$

$$y = \frac{b}{1+b} \varepsilon \quad (4)$$

Assuming that there is only one player in the system (one policy authority, namely the monetary authority) and no uncertainty in transmitting the effect of monetary policy. In this regard, k is defined as economic factors that affect the achievement of suboptimal unemployment (socially), for instance tax and insurance. And $k > 0$ reflects the central bank's desire to correct the distortion.

12.3.1. Political Transparency: Public Uncertainty concerning the Preferred Weight of Output Stabilization (Parameter b)

As explained previously, the central bank's loss function is formulated as follows:

$$L = \frac{1}{2} E[a\pi^2 + b(y - k)^2] \quad (5)$$

Departing from the Rogoff restriction, where $k > 0$, Demertzis and Hallett also suggested that perfect (full) political transparency is not only conditioned by policy preference weight in the loss function of the central bank (a and b) explicitly, but also the relative weight of both parameters (a/b) as well as the Marginal Rate of Substitution, including ($a\Delta\pi/b\Delta y$).

The issue of transparency appears when public perception of the central bank's preferences, for example, the weight of output stabilization (β), differs from the weight set by the central bank. Consequently, there is a false perception (η), considering $\beta = b + \eta$, where $E(\eta) = 0$ and $V(\eta) = \sigma_\eta^2$. This shows that in general, the (average) public perception is correct but not in specific cases. The consequence of uncertainty regarding the value of b is uncertainty in terms of the relative role

of controlling inflation by the central bank in the eyes of the public. By assuming that $a + b = 1$, then $\alpha = a - \eta$. Therefore, the relative weight of the central bank's preferences is formulated as follows:

$$\frac{\beta}{\alpha} = \frac{b + \eta}{a - \eta} \quad (6)$$

where the possibility of full transparency requires $E\left(\frac{\beta}{\alpha}\right) = \frac{b}{a}$ or $\frac{\beta}{\alpha} = \frac{b}{a} + \xi$, where $\xi = \frac{(a+b)\eta}{a(a-\eta)}$, and

$$E(\xi) = E\left[\frac{(a+b)\eta}{a(a-\eta)}\right] = \frac{(a+b)\bar{\eta}}{a(a-\bar{\eta})} - \frac{(-a)(a+b)\sigma_{\eta}^2}{a^2(a-\bar{\eta})^2} + \frac{a^2(a+b)\bar{\eta}}{a^3(a-\bar{\eta})^3}\sigma_{\eta}^2 \quad (7)$$

We can see that the assumption, $E(\eta) = 0$, is insufficient for the realization of full transparency. In this case, full transparency requires $E(\eta) = 0$ or $V(\eta) = \sigma_{\eta}^2 = 0$.

To simplify the analysis without changing the qualitative conclusion, we then assume $a = 1$. Consequently, the problem of transparency relates more to uncertainty in the actual value of parameter b . With the perceived value of β , the public expects the central bank to set:

$$\pi = \beta k - \frac{\beta}{1 + \beta} \varepsilon \quad (8a)$$

$$\pi^e = \beta k \quad (8b)$$

based on optimization:

$$\text{Min}_{\pi} L = \frac{1}{2} E\left[\pi^2 + b(\pi^2 - \beta k + \varepsilon - k)^2\right] \quad (9)$$

produces the solution of inflation in equilibrium, namely

$$\pi^* = \frac{b(1 + \beta)k}{1 + b} - \frac{b}{1 + b} \varepsilon \quad (10)$$

The solution is in line with rational expectations, considering that $E(\pi^*) = \beta k = E(\pi^e)$. Meanwhile, assuming $E(\eta, \varepsilon) = 0$, inflation variance is formulated as follows:

$$V(\pi^*) = E[\pi^* - E(\pi^*)]^2 = \left(\frac{b}{1 + b}\right)^2 (k^2 \sigma_{\eta}^2 + \sigma_{\varepsilon}^2) \quad (11)$$

Consequently, inflation variance is a function of imperfect transparency based on public perception (σ_{η}^2). If b and k are not equal to 0, with a higher value of σ_{η}^2 (transparency failure), inflation variance would also increase, considering:

$$\frac{\partial V(\pi^*)}{\partial \sigma_{\eta}^2} = \left(\frac{b}{1 + b}\right)^2 k^2 > 0 \quad (12)$$

Meanwhile, average inflation behaves independently to σ_η^2 , hence:

$$\frac{\partial E(\pi^*)}{\partial \sigma_\eta^2} = 0 \quad (13)$$

by searching for further solutions, the output equilibrium solution is:

$$y^* = \frac{\varepsilon - \mu k}{1 + b} \quad (14a)$$

$$\text{and} \quad E(y^*) = 0 \quad (14b)$$

Meanwhile,

$$V(y^*) = \frac{\sigma_\varepsilon^2 + k^2 \sigma_\eta^2}{(1 + b)^2} \quad (15)$$

The results imply that the average level of output equilibrium is unaffected by changes in the degree of transparency, while output variance would reduce in line with increasing transparency, considering:

$$\frac{\partial V(y^*)}{\partial \sigma_\eta^2} = \frac{k^2}{(1 + b)^2} > 0 \quad (16)$$

Another important implication from the solutions above is the relative effect of imperfect transparency on inflation and output stability, by comparing Equations (12) and (16). We can conclude that if the central bank behaves conservatively, namely is $b < a = 1$, then:

$$\frac{\partial V(y^*)}{\partial \sigma_\eta^2} > \frac{\partial V(\pi^*)}{\partial \sigma_\eta^2} \quad (17)$$

Therefore, imperfect political transparency will have a more adverse impact on output stability compared to inflation variability.

12.3.2. Economic Transparency: Public Uncertainty to Changes in Economic Factors (value of parameter k)

Like political transparency, in the context of economic transparency, the public have a different perception of the target set by the central bank, k . The public perceive the target as $c = k + v$, where $E(v) = 0$ and $V(v) = \sigma_v^2$. In this case, full transparency requires $E(v) = 0$ or $V(v) = \sigma_v^2 = 0$. The public expects the central bank to specify:

$$\pi = \beta c - \frac{\beta}{1 + \beta} \varepsilon \quad (18a)$$

$$\pi^e = \beta c \quad (18b)$$

Therefore, the optimization solution of the central bank's loss function is as follows:

$$\text{Min}_{\pi} L = \frac{1}{2} E[\pi^2 + b(\pi^2 - \beta c + \varepsilon - k)^2] \quad (19)$$

Where the solution of inflation in equilibrium is as follows:

$$\pi^* = \frac{b(bc + k)}{1 + b} - \frac{b}{1 + b} \varepsilon \quad (20)$$

Considering that $E(c) = k$, then the solution is also in line with rational expectations theory because $E(\pi^*) = bk = E(\pi^e)$.

Meanwhile, inflation variance is as follows:

$$V(\pi^*) = E[\pi^* - E(\pi^*)]^2 = \left(\frac{b}{1 + b} \right)^2 (b^2 \sigma_v^2 + \sigma_\varepsilon^2) \quad (21)$$

Consequently, inflation variance is directly proportional to variance v and, therefore, inversely proportional to transparency.

$$\frac{\partial V(\pi^*)}{\partial \sigma_v^2} = \left(\frac{b}{1 + b} \right)^2 b^2 > 0 \quad (22)$$

Then, output equilibrium is as follows:

$$y^* = \frac{\varepsilon - bv}{1 + b} \quad (23)$$

With $E(y^*) = 0$, we can conclude that average output is independent of σ_v^2 . Meanwhile, output variance is:

$$V(y^*) = \frac{\sigma_\varepsilon^2 + b^2 \sigma_v^2}{(1 + b)^2} \quad (24)$$

Therefore:

$$\frac{\partial V(y^*)}{\partial \sigma_v^2} = \frac{b^2}{(1 + b)^2} > 0 \quad (25)$$

The solutions above imply that economic transparency is the same as political transparency, namely that if the central bank acts conservatively, if $b < a = 1$, then:

$$\frac{\partial V(y^*)}{\partial \sigma_v^2} > \frac{\partial V(\pi^*)}{\partial \sigma_v^2} \quad (26)$$

Therefore, imperfect economic transparency would have a more adverse impact on output stability rather than inflation variability.

12.3.3. Implications of Imperfect Transparency

From the theoretical model developed above, we can draw a number of conclusions or implications from the presence of imperfect transparency as follows.

First, imperfect transparency, political and economic, does not lead to biases in terms of inflation or output. Therefore, public inflation and output expectations will align with actual conditions.

Second, in the short run, the public believes that the central bank will behave more conservatively than the actual conditions ($\beta < b$ and/or $c < k$), with the temporary possibility of experiencing lower inflation and higher output than the respective trends and vice versa for a more liberal central bank.

Third, imperfect transparency, politically and economically, influences inflation and output. In this case, for a conservative central bank, less transparency would exacerbate inflation and output variability; with a more pronounced impact on output variability and vice versa for a more liberal central bank.

Fourth, considering average inflation and output are not affected, and the ratio of output variance to inflation variance is the same for both types of transparency (political and economic), hypothetically there is no difference in the results when testing transparency models. Different results may be produced by the data/information used in the testing.

Fifth, the results of empirical testing in several OECD member countries found evidence supporting the conclusion that average inflation and output were not influenced by the degree of transparency. Nonetheless, differing degrees of transparency could explain approximately 50% of inflation variability, while the linkages between transparency and output variability were not as clearly defined.

12.4. Practices and Empirical Studies of Transparency and Communication Strategies in Different Countries

As mentioned in Chapter 2, central banks have increased monetary policy transparency over the past decade. In addition to the three salient factors described previously, namely greater central bank independence, the proliferation of ITF implementation and financial market advancement, such developments have also been advocated by international institutions through various endeavors, including promoting monetary and financial policy transparency since 1999.

12.4.1. Assessing Transparency by the IMF

Driven by the importance of policy transparency to minimize the risk of a crisis, such as those that have struck Asia and Latin America, the IMF initiated and developed transparency guidelines for monetary and financial policies, commencing in 1999, as the “Code of Good Practices on Transparency in Monetary and Financial Policies (MFP),” to which many members now adhere (IMF, 2003).

Participation, in terms of implementing the transparency guidelines, is voluntary and began with 23 member countries in the year 2000, subsequently increasing to 57 of the 184 IMF members. Four important aspects of policy transparency

were evaluated, namely: (1) clarity regarding the role, jurisdiction, and purpose of the monetary policy authority; (2) disclosure concerning the monetary policy formulation and reporting process; (3) public availability of monetary policy information; and (4) monetary authority accountability and integrity.

The assessment revealed that, in 2000, monetary policy was most transparent in terms of clarifying the role, jurisdiction and purpose of the monetary policy authority. In nearly all participating countries, the final goal of monetary policy, the duties of the central bank and authority to utilize monetary policy instruments were stipulated and regulated in prevailing laws. Central bank accountability and integrity, however, showed a number of key weaknesses. Based on that assessment, the IMF recommended several improvements to the general scope and forms of transparency (for instance, the quality and timeliness of information, publication frequency, etc.) and identified the weaknesses in terms of guaranteeing central bank accountability and integrity.

Subsequently, the assessment conducted in 2003 showed that despite remaining high, monetary policy transparency had declined on conditions in 2000. Congruent with the IMF (2003), this could be explained by several factors. *First*, the assessment in 2000 included several countries as pilot projects employing self-assessment techniques that were typically less stringent than the direct evaluation performed by the IMF itself. *Second*, a number of low-income countries were included in the 2003 assessment, where monetary policy is often less transparent. *Third*, IMF experience in terms of assessing aspects of transparency and the countries involved also improved. Consequently, the quality of the assessment in 2003 was better than in 2000.

The assessment in 2003 showed that public availability of information on monetary policy was the most transparent and exceeded the achievements of 2000. Such developments reflected the growing number of communication channels exploited by central banks to communicate with the public, not merely relying on periodic publications but also uploading data and information to websites to expand public access to information. Nevertheless, the assessment in 2003 also revealed a number of weaknesses, particularly in terms of transparent practices at central banks. Similar to the previous assessment, central bank practices and integrity remained the weakest link of monetary policy transparency.

A cross-border study on monetary policy transparency was conducted by Sundarajan et al. (2003) based on the results of a survey disseminated to 160 central banks of the 178 IMF members. The participation rate of the survey was comparatively high, with 75% of the surveyed central banks completing the questionnaire, representing a broad range of economic and political development. Transparency was assessed based on various elements of the four transparency aspects contained in the MFP Transparency Code. Specifically, the study proposed an index based on the average implementation of four monetary policy transparency dimensions, namely the means, timeliness, periodicity as well as form and content of the information submitted.⁴

⁴For a more detailed description of the method used to calculate the index, refer to Appendices I and II in Sundarajan et al. (2003).

The degree of transparency per the country classifications based on economic development and political conditions is presented in [Table 12.1](#). The table shows that monetary policy is more transparent in higher-income countries, including OECD members and advanced countries. Transparency concerning the role, jurisdiction, and purpose of the monetary policy was adequate, however, in most countries.

Most countries displayed weaknesses in terms of monetary policy authority accountability and integrity. Conspicuous differences were found, however, in terms of disclosing the policy formulation process and public availability of information on monetary policy, which are typically lower in low-income and developing countries.

12.4.2. Practices at Several Prominent Central Banks

Blinder et al. (2001) evaluated monetary policy transparency practices at several central banks located in advanced countries, specifically the Federal Reserve, ECB, BOJ, Bank of England, Bank of Canada (BoC), Reserve Bank of New Zealand (RBNZ), and Swedish Riksbank. Aspects of transparency were evaluated, including economic forecasts and the direction of monetary policy, information disclosure concerning monetary policy decisionmaking, as well as communication with and accountability to parliament.

All central banks in advanced countries publish periodic information regarding monetary policy and economic developments, encompassing economic and monetary policy reports as well as macroeconomic projections. In general, such information is reported on a quarterly and monthly basis or as required, while the macroeconomic forecasts are published in brief. Only the RBNZ and Swedish Riksbank publish detailed macroeconomic projections.

Variations were found between the seven central banks, however, in terms of publishing information on monetary policy decisionmaking. In general, the central banks in question announced their interest rate decisions immediately after the decision had been made, accompanied by an explanation of the considerations underlying any changes in the decisions taken. There were differences, however, in the publication of records from the monetary policy meetings held at the central banks, including the minutes and verbatim records. The Fed, BoJ, BoE, Swedish Riksbank, and RBNZ published their minutes, while the other central banks did not.

On the other hand, prevailing laws in advanced countries required the respective central bank to submit a monetary policy report to parliament, albeit at different frequencies ([Table 12.4](#)). Likewise, the central banks were required to provide clarification to parliament periodically, through a, generally open, plenary hearing or commission, by the Governor of the Central Bank or other senior official. In general, however, parliament or the commission were not authorized to appraise central bank performance or dismiss the Governor and/or members of the Board of Governors for the decisions taken.

Table 12.1: Monetary and Financial Policy Transparency Index by Country.

	Implementing All Elements of the MFP Transparency Code	Clarity Concerning the Role, Jurisdiction, and Goal of the Monetary Policy Authority	Monetary Policy Formulation, and Reporting Process Disclosure	Public Availability of Information on Monetary Policy	Central Bank Accountability and Integrity
Stage of economic development					
• Low income	72.0	80.9	58.6	60.2	59.6
• Low-middle income	74.6	81.0	59.9	66.4	57.8
• Upper-middle income	78.0	82.1	63.9	72.4	56.2
• High income, non-OECD	73.9	86.9	57.1	61.9	58.2
• High income OECD	92.1	90.3	88.6	84.6	66.7
Economic/political characteristics					
• Advanced	90.4	90.2	85.0	82.5	66.1
• Developing	72.6	80.4	55.6	64.7	57.5
• Transition	81.3	85.8	78.5	66.7	60.2
Total	77.6	83.3	65.3	68.6	59.8

Source: Sundarajan et al. (2003).

12.4.3. Monetary Policy Transparency and Economic Performance

As a contemporary issue, most of the literature available concerning monetary policy transparency focuses on the theoretical analysis, as reviewed in Section 12.2. Not many studies explore the impact of monetary policy transparency on economic performance. The few studies available focus on the financial markets using time series data. Nonetheless, Chortareas, Stasavage, and Sterne (2002) conducted an empirical study concerning the impact of monetary policy transparency on inflation and output using data from 87 countries. Their methodology and results are described as follows.

12.4.3.1. Methodology. In their study, Chortareas et al. (2002) focused on the publication of economic forecasts by the central bank and its impact on inflation and output volatility. The hypothesis constructed from the nascent theories developing in the literature revealed two important aspects. *First*, that publishing central bank forecasts improved inflation performance, considering that a greater degree of transparency increased the sensitivity of the central bank's reputation to the actions taken, which therefore reduced the incentives for the central bank to implement inflationary policy. *Second*, that publishing central bank forecasts reduced output variability, considering that with a greater degree of transparency, the central bank underscored the achievement of low inflation and, therefore, limited its ability to stabilize the economy, for instance, using surprise policies to overcome demand-side shocks in the economy.

The study utilized the results of a survey conducted by Fry, Julius, Mahadeva, Roger, and Stone (2000) of central banks and monetary authorities from 94 countries. The survey questioned various aspects of transparency, such as the publication and clarification of macroeconomic forecasts, explaining the monetary policy formulation process, analysis quantity, research activity and the speeches of central bank officers. The survey also assessed various aspects of accountability, independence, and target setting, which contribute to monetary policy framework clarity and transparency. Chortareas et al. (2002) emphasized the publication and clarification of macroeconomic projections by the central bank as the aspect most closely related to various other studies on monetary policy transparency.

Most of the central banks surveyed (79% of 94 central banks) published some form of forecast or forward analysis, although the form and scope varied. Most clarified the direction of the economy, such as economic growth and its components, inflation, interest rates, exchange rates, and monetary aggregates. Some of the central banks published money supply targets as part of their forward analysis as intermediate targets rather than operational targets. Some also provided a more detailed explanation of the risks contained in the forecasts and the inaccuracies found in previous projections. The profiles of the surveyed central banks, reviewed in terms of the types of forecasts published and quality of the forward analyses, are presented in [Table 12.2](#).

To measure the degree of central bank transparency in terms of publishing economic forecasts, Chortareas et al. (2002) applied a Guttman scale. Fundamentally, the index applied a dichotomous format to four questions concerning

Table 12.2: Economic Forecast Publication Profile of Central Banks.

Questions	Categories of Answers	All Countries	Industrial	Transition	Developing
Form of publication of forecasts	Words and numbers	35	16	5	14
	Either words or numbers	25	8	6	11
	Unspecified	13	0	4	9
	None	21	4	7	10
Forward-looking analysis in standard bulletins and reports	More than annually	39	18	7	14
	At least annually	24	4	4	16
	Unspecified	10	2	4	4
	Otherwise	21	4	7	10
Discussion of past forecast errors	Yes	21	8	3	10
	Sometimes	9	7	2	0
	No	64	13	17	34
Risks to forecast published	Words and numbers	9	7	2	0
	Either words or numbers	23	9	4	10
	None	62	12	16	34

Source: Chortareas et al. (2002, p. 103).

forecast publication transparency. Therefore, the index would produce a higher value for a central bank that practices all four aspects of transparency mentioned in the survey. The index of central bank transparency in terms of publishing economic forecasts based on the Guttman scale is presented in Fig. 12.1. Of the 82 countries assessed, 25 produced a Guttman scale of 0, while eight had a scale of 1, 24 had a scale of 2, six had a scale of 3 and 19 had a scale of 4.

The effect of transparency on inflation and output variability was analyzed in terms of the correlation to each transparency question. To control the effect of other variables, transparency was also analyzed using a regression model that included the influence of GDP, degree of economic openness, exchange rate system, announcement of the inflation and money supply targets, and degree of political instability. In addition, to analyze the effect of economic, political and legal conditions, the impact of transparency was measured using indicators of democracy, GDP per capita, central bank independence, and central bank analysis quality.

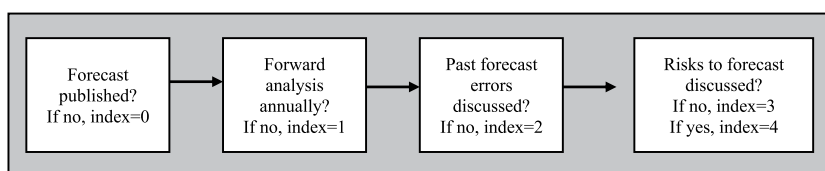


Fig. 12.1: Measuring the Transparency Index Using the Guttman Scale.

Source: Chortareas et al. (2002, p. 105).

(1) *Transparency and inflation:*

Correlation between the Guttman scale for the four transparency questions and the publication of economic forecasts by the central bank is presented in Table 12.3. From the table, we can see that inflation correlates inversely and significantly with all transparency indexes. This is empirical evidence that a greater degree of monetary policy transparency could control inflation. The strongest correlation was recorded for transparency in terms of publishing forecasts and clarifying the risks contained within.

Regression analysis further confirmed the empirical evidence that transparency could improve inflation performance.⁵ A stronger impact was recorded for countries that implemented a flexible exchange rate system. In such countries, where inflation began in double digits at around 12% per annum, the central bank's decision to publish routine inflation forecasts (a bump in the index from 0 to 2) triggered disinflation of around 1.8%–7% per annum. Conversely, in countries that implemented a fixed exchange rate system, the effect of transparency on disinflation was minimal, with inflation moderating slightly from 12% to 11.8% per annum.

(2) *Transparency and output volatility:*

The correlation between transparency and output volatility was also analyzed using the standard deviation of GDP, as presented in Table 12.4. From the table, we can see that the correlation between transparency and output volatility is generally negative, especially in terms of quarterly output volatility, but with a comparatively low level of significance. Although this analysis fails to provide strong evidence that transparency could reduce output volatility, there is also no evidence that greater monetary policy transparency would exacerbate output volatility. The regression analysis of the transparency index against output volatility, including the variable, terms of trade, also failed to provide conclusive evidence that transparency correlated with output volatility.

⁵Refer to Chortareas et al. (2002, p. 107). In addition to the transparency index, the variables log GDP per capita, openness, political instability, dummy exchange rate peg (model 1) were included in the regression model, along with pegxtransparency (model 2) as well as inflation targetxtransparency and money targetxtransparency (model 3).

Table 12.3: Transparency and Inflation: Correlation Analysis.

	Log Inflation
Guttman scale of transparency	-0.37 ($p < 0.01$)
Publication?	-0.29 ($p = 0.01$)
Forward analysis at least annually?	-0.15 ($p = 0.15$)
Past forecast discussed?	-0.21 ($p = 0.05$)
Risks to forecast considered?	-0.28 ($p = 0.01$)
Number of observations	87

Source: Chortareas et al. (2002, p. 105).

Table 12.4: Transparency and Output Volatility: Correlation Analysis.

	Standard Deviation of Annual GDP Growth	Standard Deviation of Quarterly GDP Growth
Guttman scale of transparency	-0.08 ($p = 0.47$)	-0.29 ($p = 0.13$)
Publication?	0.06 ($p = 0.59$)	-0.10 ($p = 0.60$)
Forward analysis at least annually?	0.02 ($p = 0.86$)	-0.25 ($p = 0.19$)
Past forecast discussed?	-0.22 ($p = 0.06$)	-0.20 ($p = 0.29$)
Risks to forecast considered?	0.09 ($p = 0.43$)	0.16 ($p = 0.40$)
Number of observations	76	29

Source: Chortareas et al. (2002, p. 108).

12.4.4. Transparency and the Communication Strategy in a Crisis Period

ITF implementation in the early 1990s provided sufficient impetus for a change in the way central banks communicated monetary policy to the public, becoming more open, transparent and accountable. Sound communication was believed to strengthen monetary policy effectiveness by creating anticipative markets. The GFC provided an important lesson that monetary policy communication alone was insufficient and must be integrated with other (macroprudential) policy communications to maintain FSS.⁶ In other words, monetary policy communication must adjust to prevailing financial system dynamics.

⁶Conceptually, monetary policy is a prudential regulatory instrument to create and maintain financial system stability in general, not the soundness of individual financial institutions. Analogously, microprudential policy is a prudential regulatory

This is crucial and simultaneously an onerous challenge. Delivering a “message” to the markets about the dangers of growing imbalances in the financial sector when economic conditions are good is difficult because such messages are very unpopular against a backdrop of upbeat market players. Similarly, the monetary policy response of raising interest rates in the absence of inflationary pressures is economically and politically difficult to accept because the central bank may be considered a threat to growth and the public interest.

Consequently, persuasive communication with the public regarding the importance of maintaining macroeconomic stability in the long run is also crucial. In that context, the communication strategy used under normal conditions cannot be applied when the market players are overly optimistic. Not to mention, high future economic uncertainty, particularly at the turning point of the economic cycle, represents a challenge in and of itself in terms of policy communication.

There are several reasons why communicating issues relating to FSS is more difficult than communicating monetary stability as follows:

- 1) The tools used to diagnose systemic risk are underdeveloped and not well established. Systemic risk is difficult to detect because the signals are not usually clear and the risks gradually accumulate during a boom period. Consequently, FSS policy is oftentimes taken without the support of adequate empirical evidence.
- 2) The definitions of FSS and relating policies are very broad,⁷ thus a single quantitative measuring tool does not exist that is easy to understand and simple to communicate.
- 3) The responsibility for maintaining FSS is usually divided across several institutions within a country and even across borders.

The responsibility is divided, hence making it more difficult for the public to assess whether the central bank has been successfully maintaining FSS or not. Consequently, it is better for the central bank to clearly communicate its assessment of systemic risks, the policies taken to mitigate the risks and a subjective evaluation of policy effectiveness.

Preparing the materials to communicate FSS itself is a fine balancing act.⁸ The central bank must be able to deliver salient and relevant information without

instrument to maintain the soundness of individual financial institutions. “Macroprudential policy seeks to develop, oversee and deliver an appropriate policy response to the financial system as a whole. It aims to enhance the resilience of the financial system and to dampen systemic risks that spread through the financial system” (G-30).

⁷“To limit systemic risk – the risk of disruption to financial services caused by impairment of all or parts of the financial system – that can have serious negative implication for the economy.”

⁸Since the mid-1990s, a number of central banks have published a Financial Stability Report (FSR), with the overarching goal of raising public awareness concerning financial market issues. During the approach to the global financial crisis of 2008/2009,

creating panic or triggering speculation. Furthermore, the central bank must also coordinate closely with other relevant institutions to avoid any conflicts in terms of the information delivered. When communicating FSS, “speaking with one voice” is highly recommended. A regulatory mechanism is required, therefore, to minimize conflicts, particularly during a crisis.

Meyersson and Karlberg (2012) stressed that communication is an important policy tool or instrument for the central bank to prevent and resolve a crisis, specifically in terms of providing information on the current risk assessment and the measures required to mitigate said risks. Effective communication could also lessen the consequences of a financial crisis and even avert a potential crisis from occurring. In contrast, miscommunication could contribute to a crisis or exacerbate an existing crisis. Therefore, during a crisis, the central bank is recommended to build confidence and prevent significant miscommunications, including insufficient communication, not talking in simple terms as well as not coordinating with other partners of financial stability.

Preparing the groundwork for communication, clarifying which measures will be taken as well as satisfying demand for meetings with the public are important elements of a successful communication strategy to build public confidence during a crisis. During such situations of ubiquitous uncertainty, silence is not an option because a lack of communication could sow the seeds of public discontent toward the policy authority. Moreover, the policy authority cannot remain silent and must provide the uninformed public enough information to fill the void without speculating on the potential risks and implications, which would surely compound the unfolding crisis. Nevertheless, full transparency could also be counterproductive if it leads to public panic. Furthermore, pre-emptive communication is critical. Clear communication regarding the problems faced and future measures to be taken would allow the central bank to build credibility and restore public confidence, even before the measures are implemented.

During a crisis, the communication strategy must be tailored to the needs of the target audience, depending on the characteristics and implications of the crisis that is unfolding. There are target groups who require direct communication channels with the central bank. One method used by the Riksbank and other central banks (including the Fed and BoE) is to post questions and answers on their respective websites. In addition, during a crisis period, statements from the central bank through speeches and interviews are also effective in terms* of swaying the markets because communications as reports/publications (for instance, the Financial Stability Report – FSR) are usually released per a fixed schedule, while speeches and interviews can be delivered at more irregular times, making the markets wait and react to the information contained within. Consequently, communication through speeches and interviews is more useful during a crisis.

Finally, to prevent conflicting messages that can seriously distort public confidence during a crisis, a centralized and internally coordinated communication

the focus of FSR shifted towards efforts to identify and assess potential significant risks to financial system stability.

strategy is recommended, bolstered by coordination with other financial stability authorities. As mentioned previously, speaking with one voice is also recommended during a crisis, thus requiring a regulatory mechanism to minimize conflicts.

In conclusion, during a crisis episode, the central bank must: (1) communicate clearly and directly about the potential risks that could arise; (2) prepare an appropriate policy response if the risks materialize; and (3) prepare a communication plan to publically clarify the policy actions required in a clear and timely fashion.

12.5. Transparency and the Communication Strategy in Indonesia

12.5.1. Transparency to Strengthen the Monetary Policy Framework

The importance of transparency and accountability was stipulated in Act No. 23 of 1999 when Bank Indonesia gained greater independence. The role, authority and goals of Bank Indonesia's monetary policy are already contained within the Bank Indonesia Act. The monetary policymaking and formulation process through the Board of Governors meeting (RDG) has also clearly been stated, along with Bank Indonesia's relationship to the Government. Furthermore, in accordance with Article 58 of the Bank Indonesia Act, Bank Indonesia is required to deliver transparent public information through the mass media at the beginning of each financial year containing: (1) an assessment of monetary policy implementation in the previous year; and (2) a monetary policy plan and targets for the upcoming year, taking into consideration the inflation target as well as economic and financial developments. Several regulatory aspects of Act No. 23 of 1999 are consistent with the basic provisions of monetary policy transparency suggested by Blinder et al. (2001) and contained in the "Code of Good Practices on Transparency in Monetary and Financial Policies" in development by the IMF since 1999 and now widely practiced by IMF member countries (IMF, 2003).

Transparency and accountability at Bank Indonesia were further emphasized through an amendment to the Bank Indonesia Act, namely Act No. 3 of 2004. Per the amendment, Article 58 stipulates that Bank Indonesia is accountable for task implementation to the People's Representative Council, whereas before the amendment the emphasis was to the public at large. The reports submitted before the People's Representative Council (and also the Government) at the beginning of each year contain: (1) task and authority implementation in the previous year; and (2) the policy plan, targets as well as task and authority implementation for the upcoming year, paying due consideration to inflation as well as economic and financial developments. Besides, Bank Indonesia also reports its task implementation and authority before the People's Representative Council and Government on a quarterly basis. The annual and quarterly reports are subsequently used to assess the performance of the Board of Governors and Bank Indonesia as a whole.⁹

⁹Demand for accountability increased after the Supervisory Board was established. The Supervisory Board has access to Bank Indonesia's budget; therefore, the Board

It is not enough to merely submit the report on task implementation, the policy plan and the policy targets to the People's Representative Council and Government, the public (and markets) must also be kept informed. It is important for the public to know the policy direction pursued, while such measures also discipline Bank Indonesia to achieve the inflation target, thereby increasing Bank Indonesia's credibility.

Bank Indonesia has implemented several measures to improve monetary policy accountability and transparency. At the beginning of each year, after discussing and deciding at the Board of Governors meeting, Bank Indonesia publically clarifies its evaluation of monetary policy performance in the previous year and announces the inflation target and orientation of monetary policy for the upcoming year. This is done through press conferences, supplements published in the mass media as well as through discussions with academia, the banking industry and business community. Formal reports concerning the evaluation and direction of monetary policy are also submitted to the People's Representative Council as material to appraise Bank Indonesia and BoG performance, as well as to the government as information. Bank Indonesia also publishes an annual report containing assessments of the economy and the monetary policy direction pursued.

Each quarter, Bank Indonesia also communicates its assessment of inflation and economic dynamics as well as the orientation of monetary policy to be pursued in the subsequent period through press conferences, supplements published in the mass media and, if necessary, Focus Group Discussions (FGD). In accordance with prevailing laws, Bank Indonesia also reports its task implementation and jurisdiction to the People's Representative Council, containing an assessment of inflation and economic conditions as well as monetary, banking and payment system policy. The report is also submitted to the Government as information and published in order to increase transparency. In terms of a monthly review, the decisions taken at the Board of Governors meeting are communicated to the public through the Monetary Policy Review, published monthly.

On top of reporting before the People's Representative Council and publishing regular statements, Bank Indonesia is also tasked with increasing monetary policy transparency, which is achieved through seminars, workshops, and other kinds of communication media provided to academia, the mass media and public to clarify the latest economic developments and monetary policy instituted by Bank Indonesia. Furthermore, Bank Indonesia also cooperates with various universities to include the range of policies implemented by Bank Indonesia, including monetary, banking and payment system policies, as well as the relevant theories, concepts and rationale into the curriculum to enrich student understanding concerning the practical application of the theories taught.

can question policy aspects. The duties of the Supervisory Board include: (1) reviewing the annual financial statements of Bank Indonesia; (2) reviewing the operating budget and Bank Indonesia intervention; and (3) reviewing the operational decision-making procedures beyond monetary policy and asset management at Bank Indonesia.

In terms of the regulatory framework contained in prevailing laws and the practices implemented by Bank Indonesia, monetary policy transparency in Indonesia is consistent with the principles and norms proliferating internationally. In fact, several aspects have been applied in Indonesia beyond those practiced elsewhere, even in advanced countries. This is specifically the case in terms of the People's Representative Council appraising the performance of Bank Indonesia and the Board of Governors pursuant to Act No. 3 of 2004, which is not regulated or possible in other countries, with the only exception of New Zealand, where the contract between the Governor of the Central Bank and Minister of Finance is regulated. Another key difference lies in the publication of Board of Governors meeting minutes, for which there is no consensus among other countries or uniform best practices agreed.

12.5.2. Bank Indonesia's Communication Strategy

The development of Bank Indonesia's communication strategy is inextricably tied to Bank Indonesia's evolution as an institution. Enactment of the new Bank Indonesia Act in 1999 extricated Bank Indonesia from government control, giving the central bank independence and holding it accountable to the public. Consequently, public expectations of Bank Indonesia's performance had to be absorbed and accommodated when formulating appropriate policy to be communicated transparently.

In other words, Bank Indonesia embodies effective communication as an instrument to anchor or direct public expectations concerning the policies taken and the targets to be achieved. Communication effectiveness is as necessary as the interest rate policy response.

That fundamental change precipitated further evolution at Bank Indonesia, from an inward-looking organization to becoming an outward-looking organization. Bank Indonesia was starkly aware of the mutual dependence between Bank Indonesia and the public. Therefore, efforts were required to nurture the symbiotic relationship between Bank Indonesia and the public through open communication and disclosure. Most importantly, open communication is one of the pillars to build Bank Indonesia credibility as a central bank. In this case, the strategic role of a communication work unit is critical. Operationally, the issue of how far Bank Indonesia applies an effective communication strategy is an important topic to observe.

There are at least three indicators used as benchmarks of communication strategy effectiveness at Bank Indonesia, namely:

- (1) How far can the communication strategy transmit the decisions taken internally at Bank Indonesia?
- (2) How far can the communication strategy transmit the key messages to form positive public expectations without getting caught in the transparency trap?
- (3) How far can the communication strategy play a role in terms of issue management and issue resolution?

Regarding the first indicator, the most strategic issue that always receives attention is aligning perceptions about the short/long-term policy targets and policy direction to be pursued, primarily in the high-level structure, namely the Board of Governors. This is not easy due to the often disparate perceptions and backgrounds of the board members. Nonetheless, experience has shown that although the discussions are often distinguished by opposing arguments, there is a generally positive learning process that improves the efficiency of decisionmaking at the Board of Governors meeting over time.

In terms of the second indicator, the success of the communication strategy is measured by the ability to form and direct public expectations. In this case, publications and clarification of the inflation and macroeconomic forecasts, as well as the growth assumptions and policies instituted by Bank Indonesia, demand communication that goes beyond dialog and press conferences to include meetings with journalists, economic investors and financial market analysts as the key to forming public expectations. This can be seen from the increasingly convergent expectations of economic indicators with Bank Indonesia's own forecasts.

Meanwhile, to avoid being caught in the transparency trap, in addition to choosing which strategic key messages to convey, the communication strategy plays an active role in the public learning process, including the general public, academia and politicians. Several programs, either delivered on a case-by-case basis or indeed regular programs, have been formulated and implemented effectively. For example, in 2003, Bank Indonesia officially developed and implemented a central banking curriculum to teach institutional, policy and management aspects of central banking as a stand-alone or integrated module, possibly the first of its kind in the world. With the degree of openness measured by the application of that curriculum, monetary policy implementation by Bank Indonesia is expected to be interpreted positively by the stakeholders, which, in turn, would support effective monetary policy implementation itself.

The final aspect relates to the communication strategy in terms of issue management and issue resolution. Regarding issue management, direct public communication through press conferences is typically preceded and followed by a number of forums. At this stage, one message is conducted repeatedly. The key messages are repeated vertically for deepening and horizontally to observe the linkages with other issues. Such forums have been developed as discussions forums with observers and chief editors from the mass media as well as speakers at various seminars. To mitigate the possibility of misinterpretation, the messages are also disseminated in print format as a press release or as seminar materials.

In addition to such forums, Bank Indonesia also communicates actively through its official website, www.bi.go.id, which not only contains regular information on economic developments and monetary policy as well as the related data, but also the press releases published in Indonesian and English for public access. Furthermore, Bank Indonesia has made large investments in technology to ensure the website contains appropriate information that is accessible to the public in real time. Through such efforts, Bank Indonesia has repeatedly been recognized for maintaining the best website in the government domain.

Another critical aspect of Bank Indonesia's communication strategy is the consistent management of internal issues and communication flow originating from the central bank or other institutions. Internal issues and communication flow are relatively simpler to manage. To that end, regular meetings are stipulated in the Memorandum of Understanding (MoU) signed between Bank Indonesia and the Ministry of Finance to discuss issues relating to achievement of the inflation target.

12.6. Concluding Remarks

This chapter has shown that the proliferation of monetary policy transparency by central banks has been inevitable, especially during the past decade. The phenomenon is consistent with the spread of central bank independence in a democratic government, the implementation of ITF-based monetary policy as well as greater awareness concerning the importance of managing expectations in increasingly advanced and complex financial markets. To that end, there are a number of interesting characteristics to observe.

First, various efforts have been taken to formulate a set of best practices to be used as a reference for the central bank in response to growing demand for monetary policy transparency. The work of Blinder et al. (2001), "How Do Central Banks Talk?," in the Geneva Reports on the World Economy No. 3 as well as the "Code of Good Practices of Transparency in Monetary and Financial Policies" developed by the IMF in 1999 are central to those efforts.

Second, several theories and analyses have been developed in the literature of monetary economics to illuminate transparency and its links to monetary policy, the rationale and the impact on the economy. Fundamentally, such theories relate to the problem of asymmetric information in monetary policy, primarily in the context of the focus and clarity of the central bank's preferences concerning the goal of monetary policy, the relevance, and influence of economic expectations in response and in terms of transmitting monetary policy as well as the debate on rules versus discretion in monetary management.

Third, although limited, several studies have provided empirical evidence for the positive contribution of greater central bank transparency on economic performance. Specifically, increasing transparency could help efforts to lower inflation, as reflected by the inverse correlation between transparency and inflation. On the other hand, a greater degree of transparency could also help to reduce output volatility or, at the very least, not exacerbate the problem.

Moving forward, the phenomenon of increasing central bank transparency shall continue and rapidly become a key menu item for central banks in terms of monetary policy implementation. Central banks are still required to assess and improve monetary policy transparency and the communication strategy, referring to best practices developed internationally, the monetary policy regime adopted, and the financial market developments in the country involved. Looking ahead, several improvements are still required in terms of the scientific and theoretical studies, including the modeling approach as well as analyzing the impact on economic performance, inflation and the central bank institutionally. Empirical studies are also required to analyze the impact of transparency on economic and financial market performance.